
Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species’ visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and frontier AI systems rarely co-train encoder and memory from scratch along a sensor axis. We test this instead in a silicon analogue: a navigation agent where only the visual sensor varies along an axis of encoder spatial bandwidth, with architecture, task, and training held fixed. Across this axis, agents reach near-identical task performance, yet the recurrent-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the recurrent state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift rather than a total-information collapse. We frame this as an emergent *capacity allocation* between the visual encoder and the recurrent integration of GPS-sensor history. Behaviourally, memory transplants are asymmetric (rich-encoder policies fail to use bottleneck states, while bottleneck policies still drive themselves from rich-encoder ones), indicating policy-level consequences of the format differences. Probe-readability and policy reliance further dissociate as independent axes: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute. The pattern connects to the invariance–disentanglement reading of the information bottleneck and to the no-remapping vs. global-remapping axis of hippocampal coding under hidden-state inference. This positions sensor structure as a training-time lever on the format of cognitive maps in artificial agents, offering a content-level alternative to training-distribution accounts of vision–language spatial-reasoning failures. More broadly, this capacity allocation generates falsifiable predictions for transformer-based world models that share the encoder–memory pipeline, and suggests that bandwidth restriction may be cognition-enabling rather than merely compute-saving.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture. Vision–language models (VLMs) with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025]. The default reading attributes this to a training-distribution gap. We pursue a structural reading instead: the encoder and the memory downstream of it are a co-trained system rather than separable modules, and *the encoder’s structure shapes what is linearly readable from the memory about space* — a content-level claim about the perception–memory interface.

Cognitive neuroscience offers concrete precedent: primate hippocampal cells respond to combinations of viewed location, head direction, and self-position [Wirth et al., 2017, Rolls, 2024], in contrast to

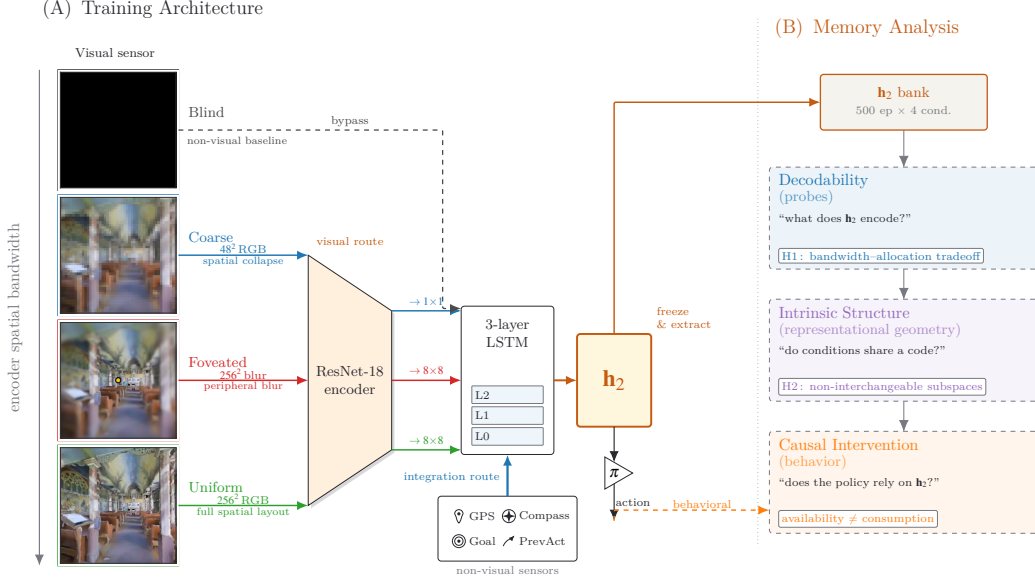


Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span an axis of encoder spatial bandwidth (*blind, coarse, foveated, uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix G) adds a 5th intermediate point on the same axis. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $h_2 \in \mathbb{R}^{512}$ to test three axes of one capacity allocation: *magnitude, format, consumption* (§3.2–3.4; synthesis in §5). Full setup in §3.1.

the more place-pure coding documented in rodents with near-panoramic vision; flying bats globally remap their hippocampal code between vision and echolocation [Geva-Sagiv et al., 2016]; and congenitally blind humans show larger hippocampal volume and supranormal non-visual spatial navigation [Fortin et al., 2008]. We treat these as motivation rather than established parallel: whether sensor structure plays a similar shaping role in artificial navigation agents is a hypothesis to test in a controlled setting rather than a parallel to demonstrate.

Cognitive-map-like representations emerge in the recurrent state of deep-RL navigation agents without being designed in, even in agents with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018].¹ This literature has typically reported one condition at a time — a fully sighted or a fully blind agent — leaving open how encoder structure interacts with what the memory builds. We adopt the same agent family as a *comparative* chassis (Figure 1): identical encoder–memory–policy pipeline, task, and training; varying only the visual sensor across four conditions plus a log-polar control that span an axis of encoder spatial bandwidth (full definitions in §3.1).

Our central empirical finding is that *rich visual encoders reduce the linear-readable GPS-derived position signal in the policy-readable top-layer hidden state*: as encoder spatial bandwidth grows, top-layer linear GPS R^2 falls monotonically, even though task performance is near-identical across conditions. We frame this pattern as an emergent *capacity allocation* between two routes carrying spatial information — the visual encoder and the recurrent integration of GPS-sensor history — with the share each route carries tracking encoder spatial bandwidth. We test this allocation along three orthogonal axes: a *magnitude* axis (how much linear position information lives in the recurrent state), a *format* axis (which subspace of the hidden state carries it across conditions), and a *consumption* axis (which route the policy actually uses for behaviour). For visual intelligence, capacity allocation offers

¹“Cognitive map” in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (§4.3, relational across rooms) and predictive-horizon analysis (Appendix E.3, SR-style future-state code in the sense of Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability probes are deferred to follow-up work.

an architecture-side complement to data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may structurally divert capacity from the downstream substrate’s spatial code rather than merely fail to teach it — and conversely, bandwidth-restricting perception (foveation, log-polar sampling) may be cognition-*enabling* for downstream spatial tasks rather than only compute-saving. For cognitive neuroscience, our setup is a controlled silicon testbed that generates falsifiable predictions about how sensor-niche structure shapes the format of cognitive maps (§5.2).

2 Related Work

This paper sits at the intersection of five lines of work: VLM spatial-reasoning failures, encoder-memory pipelines in embodied AI, foveated perception, emergent maps in deep-RL navigation, and cognitive-map / place-cell theory in neuroscience. We also draw on the interpretability methodology surveyed at the end of this section.

Spatial-reasoning failures in vision-language models Multimodal LLMs with strong visual recognition systematically fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025]. Two diagnoses compete: a *data-side* account treats the gap as a training-distribution problem [Chen et al., 2024], and an *encoder-side* account treats it as structurally upstream of downstream training — probing studies show pretrained encoders with different objectives carry markedly different geometric content, with self-supervised encoders retaining depth and surface structure that vision-language-trained ones lack [Tong et al., 2024b, El Banani et al., 2024, Tong et al., 2024a]. Our paper provides a controlled analog of the encoder-side account in a recurrent navigation setting; whether the same mechanism transfers to attention-based memory in VLMs is a prediction, not a result.

Encoder-memory interface in modern embodied AI Recent world models [Hafner et al., 2023, Bardes et al., 2024] and embodied vision-language-action agents [Driess et al., 2023, Kim et al., 2024] share an architectural pattern: a pretrained visual encoder feeds an attention- or recurrence-based memory module read by a downstream policy. Practitioner choices — multi-encoder fusion [Kim et al., 2024], geometry-driven encoder selection [Zhou et al., 2024], frozen-feature extraction [Nair et al., 2022, Majumdar et al., 2023] — already reflect that encoder content shapes downstream representations. Whether transformer-based navigators [Zeng et al., 2024] integrate sensor history the way LSTM bottleneck conditions do here is an open question.

Foveated and bandwidth-constrained vision Peripheral-vision phenomenology [Strasburger et al., 2011, Rosenholtz, 2016] motivates deep-learning work applying eccentricity-dependent processing to neural networks, mostly under a compute or recognition prior [Mnih et al., 2014, Shang and Ryoo, 2023]. Deza and Konkle [2021] are the closest precedent for treating foveation as content-shaping. Atanov et al. [2024] train PointGoalNav agents across a pixel-bandwidth spectrum without examining the recurrent state; Pourrahi and Bashivan [2025] probe a foveated visual-search model (no embodied locomotion). We extend this line by asking how sensor bandwidth reshapes the format of downstream spatial memory in a recurrent navigation policy.

Emergent maps in deep-RL navigation Map-like structure emerges in the recurrent state of navigation agents trained under absent or spatially uniform perception [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. Self-supervised path-integration can produce multi-modular grid cells [Schaeffer et al., 2023b], though grid-like decoding can be artefactual without careful controls [Schaeffer et al., 2023a]. This literature studies one condition at a time; we go comparative across the encoder-bandwidth spectrum.

Cognitive map theory and place-cell remapping The cognitive map hypothesis [Tolman, 1948, O’Keefe and Nadel, 1978] casts internal models of space as flexible navigation substrates. Modern accounts model hippocampal codes as predictive maps over future state visits [Stachenfeld et al., 2017] and relational generalisers binding sensory observations to abstract structure [Whittington et al., 2020, Behrens et al., 2018]. Place-cell rate maps shift along a remapping spectrum — no remapping (all fields shared between contexts), rate remapping (same fields, different rates), and global remapping (no shared fields) [Sanders et al., 2020] — which Sanders et al. [2020] formalise as hidden-state inference (place fields remap when sensory evidence supports distinct contexts). Comparative cognition documents how sensor structure shapes navigation across species: blind subterranean rodents path-integrate without vision [Kimchi et al., 2004], echolocating bats build

cognitive maps [Toledo et al., 2020, Geva-Sagiv et al., 2015]. We adopt this analysis vocabulary — Skaggs information [Skaggs et al., 1996], the remapping spectrum — and read the four sensor conditions as silicon analogs of distinct sensory niches.

We draw on three interpretability families: probing classifiers with selectivity controls [Belinkov, 2022, Hewitt and Liang, 2019]; representational geometry [Kornblith et al., 2019, Williams et al., 2021, Facco et al., 2017, Ansuini et al., 2019]; and causal/behavioural intervention on the recurrent state (memory transplant and shortcut discovery in this work; principled upgrades for follow-up: distributed alignment search [Geiger et al., 2024] and Sussillo–Barak fixed-point analysis [Sussillo and Barak, 2013]). Convergence across the three deployed families on the same condition-level contrast is the strongest claim our methodology supports.

3 Methods

3.1 Experimental setup

Task and training. PointGoal navigation in Habitat [Savva et al., 2019] trained with DD-PPO [Wijmans et al., 2020]. Episodes terminate on success (within 0.2 m of the goal) or 2000-step timeout; per-episode metrics use SPL [Anderson et al., 2018]. The training pool merges Gibson-0+ [Xia et al., 2018] (411 scenes) with Matterport3D-train (61 scenes); held-out evaluation uses 18 MP3D test scenes (1008 episodes) for distribution-shift probes. All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar). Uniform 250M-frame budget, single seed per condition — matching standard practice in the comparative-cognition modelling literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023b]; rigour at the cross-condition level instead comes from convergent multi-method evidence (§5.5), supplemented by an independent-retraining replication of the blind condition (Appendix A). Full hyperparameters, action space, and architecture in Appendix A.

Visual sensor conditions. Five conditions span an axis of encoder spatial bandwidth; the first four are the main bandwidth-axis points, and *foveated-logpolar* is included as a discriminating control (see end of list):

- *Blind*: no visual stream (encoder bypassed).
- *Coarse*: 48×48 uniform RGB; ResNet-18’s 32-fold spatial downsampling collapses the feature map to 1×1 , removing per-step spatial structure while preserving channel information.
- *Foveated*: 256×256 RGB under eccentricity-dependent Gaussian blur ($\sigma_{\max}=8$, quadratic falloff, gaze fixed at image centre); 4×4 encoder feature map.
- *Uniform*: unblurred 256×256 RGB; 4×4 encoder feature map.
- *Foveated-logpolar* (control): 256×256 RGB under log-polar resampling onto a 64×64 retinal grid, yielding a $\sim 2 \times 2$ encoder feature map (Appendix G). Isolates the encoder-spatial-output mechanism from blur per se and tests whether the bandwidth-axis monotonicity reading is the right account.

Probing data collection. After training all weights are frozen and we collect per-step top-layer LSTM hidden states ($\mathbf{h}_2 \in \mathbb{R}^{512}$) from $n = 500$ Gibson held-out episodes under deterministic (argmax-action) rollouts.² The probing methodology then follows the three blocks of Figure 1 (§3.2–3.4); convergence across the three is the strongest claim the methodology supports. Hyperparameters and protocol specifics are in Appendix B.

3.2 Decodability: what does $\mathbf{h}_2$ encode about position?

Position information must be in $\mathbf{h}_2$ before format claims are meaningful, and two confounds drive multi-method redundancy. First, a single static readout cannot separate linear-readable structure from memorisation (linear R^2 inflates) or from non-linear codes the policy cannot use (MLP R^2 overestimates). Second, GPS enters the LSTM as a per-step L0 input, so static decodability does not

² $\mathbf{h}_2$ is the policy-readable slice; full (L0, L1, L2) $\times$ (hidden, cell) probes are in Appendix 10. Stochastic-action rollouts collapse to ~ 4 steps via STOP entropy and are not used (Appendix B).

by itself distinguish integrated memory from passthrough; ruling out passthrough requires temporal tests passthrough cannot produce.

Snapshot decodability: what is encoded right now? Linear Ridge probes with *Hewitt-Liang label-permutation selectivity* [Hewitt and Liang, 2019] (memorisation control); 2-layer MLP probes (non-linear ceiling); MINE [Belghazi et al., 2018] for $I(\mathbf{h}_2; \text{position})$ as a probe-form-independent anchor; per-unit *Skaggs spatial information* [Skaggs et al., 1996] with a circular-shuffle null for place-unit counts.

Memory dynamics: integrated code or per-step passthrough? *Step-binned probing* (separate Ridge per log-spaced step-in-episode bin) for within-episode time-variation; *lag-k retrospective probe* (decode GPS_{t-k} from $\mathbf{h}_t$) for memory persistence; *predictive-horizon probe* (decode $(x, y)_{t+k}$) testing the successor-representation prediction of Stachenfeld et al. [2017] — positive future R^2 is impossible under passthrough since future GPS is not yet an input; *excursion-forgetting probe* (mid-episode random-action detour and recovery) as a perturbation test.

The two together locate the position code on a magnitude axis (linear/MLP/MI), a population axis (Skaggs), and a temporal axis (lag/horizon/recovery), supplying the magnitude axis, the place-cell vocabulary, and the memory-vs-passthrough distinction.

3.3 Intrinsic structure: do conditions share a code?

The format claim — conditions allocate to non-interchangeable subspaces of $\mathbf{h}_2$ — has two independent sides: *within-condition* (is one condition’s linear position axis scene-invariant or scene-conditional?) and *across-condition* (do different conditions share a basis?); both must be measured.

Within-condition: is the linear axis scene-invariant? *Leave-one-scene-out (LOSO)* cross-validation operationalised as a no-remap vs. global-remap test in the sense of Sanders et al. [2020]; cross-scene preferred-bin Spearman between rate maps as a per-unit complement.

Across-condition: do conditions share a basis? Cross-condition probe transfer (train Ridge on A , test on B); *Linear CKA* [Kornblith et al., 2019]; 1-NN purity; *generalised Procrustes shape distance* θ_1 [Williams et al., 2021], a true metric satisfying the triangle inequality (unlike CKA); principal angles between top- K PCA bases plus position-direction cosines; per-condition complexity via participation ratio and TwoNN intrinsic dimension [Facco et al., 2017, Ansuini et al., 2019].

The two together produce the format claim from two independent angles (within-condition scene-invariance dichotomy + across-condition geometric distance), jointly hard to fake by single-method artefact.

3.4 Causal intervention: does the policy rely on $\mathbf{h}_2$?

Probe-readability is necessary but not sufficient for behavioural relevance: a code can be linearly readable yet policy-irrelevant, or non-linearly encoded yet policy-relied. The two blocks above are correlational at the representation level; consumption requires causal tests outside any probe framework.

Representational causation: can foreign or stale memory be used? *Memory transplant*: mid-trajectory paste of donor $\mathbf{h}_2$ into recipient across the full 5×5 donor-recipient matrix; a self-transplant (donor from a different episode of the same condition) controls for replacement disturbance. *Memory-init transplant*: re-initialise from end-of-previous-episode memory vs. zero, isolating trajectory-bound vs. scene-generic memory.

Behavioural tests: does intact memory change behaviour? *Shortcut discovery* (Tolman 1948 in silico, paired same-scene-different-goal episodes): SPL drop under reset vs. persistent LSTM indexes how tightly the policy couples actions to integrated memory. *GPS-sensor mid-rollout ablation*: zero GPS and compass from step 0, measure end-of-episode success.³

³Zero-input ablations push the LSTM off its training distribution, so we use them only at the policy-output (success) level; an in-distribution alternative (replacing GPS/compass with training-marginal samples) is left for future work.

Table 1: Per-condition headline. All agents trained to a uniform 250M frames (§3.1); blind uses an independently retrained checkpoint (Appendix A). Probe R^2 columns: 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). The mid-episode story differs from the long-tail story: all four sighted conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200, Figure 25b); the rich-encoder long-tail collapse produces the sub-chance no-cap numbers here. Bold marks (i) blind/coarse rows where GPS+Compass remain decodable (magnitude-axis phenomenon; one-sample t -test against zero, $p < 0.001$) and (ii) best task performance. The wide σ in rich-encoder rows reflects probe-at-chance behaviour: the mean GPS R^2 is statistically indistinguishable from zero (foveated $t \approx 0.60$, $p \approx 0.58$; uniform $t \approx -1.34$, $p \approx 0.25$; one-sample t -test, $df=4$). The magnitude-axis claim is that linear top-layer readability *collapses to chance* in rich-encoder conditions, not that it is reliably negative.

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	250M	0.59	0.93	$+0.934 \pm 0.04$	$+0.844 \pm 0.03$
Coarse	250M	0.853	0.989	$+0.582 \pm 0.099$	$+0.659 \pm 0.061$
Foveated	250M	0.894	0.993	$+0.178 \pm 0.659$	$+0.503 \pm 0.142$
Foveated-logpolar	250M	0.879	0.990	-0.029 ± 0.759	$+0.467 \pm 0.330$
Uniform	250M	0.891	0.993	-1.785 ± 2.993	-1.244 ± 3.579

The two together dissociate probe-readable from policy-relied capacity; the transplant matrix and the shortcut SPL drop together populate the consumption axis of the synthesis (Figure 6). Method-family scope, single-seed protocol, and the boundaries of these intervention tests are detailed in §5.5.

4 Results

4.1 Comparable navigation skill across all five conditions

Table 1 collects the headline per-condition numbers. **All five agents learned; none failed.** The four sighted conditions reach 98–99% success and blind reaches 93%; a Bug-style controller without learning achieves SPL 0.07 on the same scenes — an order of magnitude below every trained agent (Appendix B). The narrow success bracket across conditions rules out a skill-gap account for any cross-condition internal difference below.

The five agents are shown different visual inputs but reach near-identical task competence on identical task and training. *They must therefore have learned different internal representations of the same task.* The open question the rest of §4 addresses is *what kind* of representation each condition learned: how the recurrent state encodes spatial position, how those encodings differ across conditions, and what this says about the encoder–memory race.

4.2 Magnitude axis: linear position code shrinks with encoder bandwidth

The magnitude-axis claim, precisely. Rich visual input *reduces the linear readability of the GPS-derived world-frame position signal at the policy-readable LSTM layer*: as encoder spatial output dimension grows, top-layer linear GPS R^2 falls monotonically across the four main conditions (Figure 2), with the foveated-logpolar control at $\sim 2 \times 2$ landing in the rich-encoder regime — a falsification of the simplest spatial-output-dimensionality reading and motivation for the spatial-feature-variety reframing (Appendix G). Goal-distance decodes at $R^2 \in [0.81, 0.90]$ in every condition, ruling out a probe-capacity reading. The contrast is specifically about whether a *linear* top-layer readout — the operation the DD-PPO policy head performs — can recover position, not whether position is encoded anywhere in $\mathbf{h}_2$. The blind result replicates [Wijmans et al., 2023]; coarse extends it to a condition with visual input but 1×1 encoder output, suggesting the shared trigger is minimal visual-feature variety per step, not absence of vision. Throughout this section “GPS code” is shorthand for “linearly decodable top-layer GPS-derived position signal”.

Information allocation: linear collapse vs. MLP recovery. The format-shift-not-info-collapse claim is sharpened by three measurements (Figure 3, Appendix E.4). Linear (Ridge $\alpha = 10$) GPS R^2 varies across a ≈ 2.1 -magnitude range over the five conditions ($+0.94$ blind to -1.19 uniform). A 2-layer MLP probe compresses this to ≈ 0.65 ($+0.99$ blind to $+0.48$ uniform), recovering $R^2 \in [0.51, 0.73]$ in rich-encoder conditions where linear is at chance; the linear–MLP gap widens

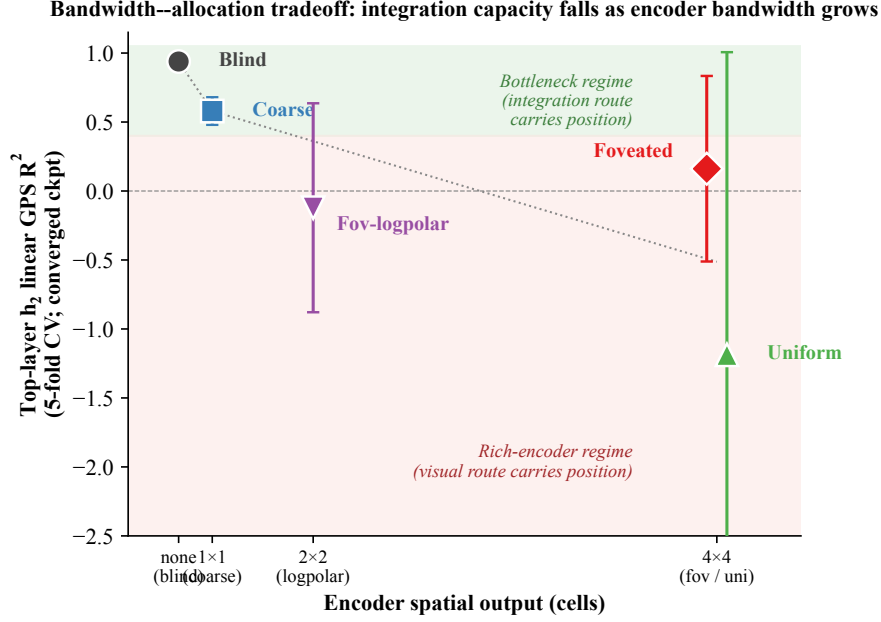


Figure 2: **The capacity-allocation principle, quantitative form.** Top-layer h_2 linear GPS R^2 (5-fold CV mean $\pm \sigma$) plotted against encoder spatial output dimension. All sighted conditions at 250M frames; blind at 340M (Appendix A). Conditions: blind (no encoder), coarse (1×1), foveated-logpolar ($\sim 2 \times 2$, Appendix G), foveated (4×4 blurred), uniform (4×4). The monotonic anti-correlation is the bandwidth-allocation tradeoff: as encoder spatial bandwidth grows, less integration capacity remains in the recurrent state. Foveated and uniform sharing x but differing in y shows the design space is not 1-D — channel-level information is a second axis, exploited in §4.5.

monotonically with encoder bandwidth. MINE estimates of total $I(h_2; \text{pos})$ shift only $\approx 1.3 \times$ across the same range. Position information is approximately preserved by non-linear readout while linear-readable information collapses — a format shift, not a total-information loss. Since the DD-PPO action head is a linear projection of h_2 , “linearly decodable at L2” maps directly to “policy-accessible”.

Temporal dynamics and capacity reallocation. We propose this reflects a *capacity-allocation* mechanism: the L0 GPS-sensor input integrated across time and the encoder’s position-correlated visual features are two routes competing for finite hidden-state capacity. Bottleneck conditions force capacity to integration; rich-encoder conditions provide a viable visual route and policy gradient reallocates capacity to it. Two within-episode and cross-training observations support this. *Within-episode* (step-binned probing, Figure 25b): rich-encoder conditions have R^2 of 0.6–0.8 in the typical mid-episode window (steps 50–200) but drop to chance in the long-tail (steps 1024+); the lag- k retrospective probe corroborates ($R^2 \geq 0.89$ for blind across $k = 0\text{--}20$, with rich-encoder profiles unstable, Appendix B). *Across training* (Figure 26): all four sighted conditions show $R^2 \approx +0.91$ at ~ 50 M frames, then the three rich-encoder conditions decay along a trajectory whose timescale tracks encoder informativeness (uniform fastest, foveated/foveated-logpolar slower) while coarse preserves $R^2 \in [0.43, 0.58]$ throughout. Compass codes follow the same pattern (Figure 26b). The pipeline-view decomposition (Figure 25c) localises the divergence: the encoder is at chance for all sighted conditions; the L0 jump is the GPS-sensor embedding entering the LSTM; the bottleneck-vs-rich-encoder split emerges specifically at L2.⁴

Population coding and perturbation evidence. At the population level (Figure 27), per-unit Skaggs spatial information [Skaggs et al., 1996] is essentially condition-flat (1.16–1.39 bits, 0.23–

⁴The single-seed cross-dataset shift on MP3D — foveated GPS at chance $\rightarrow +0.35$ — awaits multi-seed replication. The MLP result alone cannot exclude exploitation of position-correlated features (Appendix C). A representational causal probe (mid-rollout encoder-output zeroing, which severs the visual route in-manifold) is left for future work.

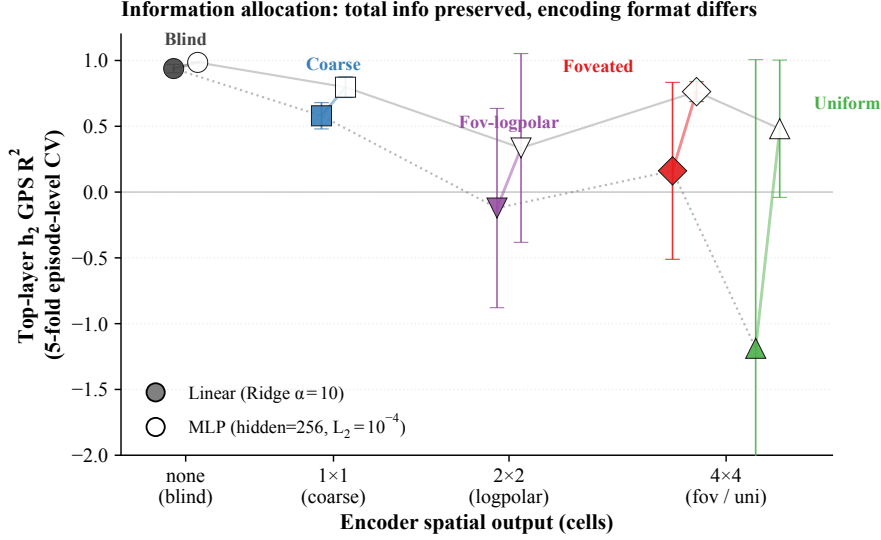


Figure 3: **Information allocation: linear vs. MLP readout.** Per-condition top-layer $\mathbf{h}_2$ GPS R^2 under linear (filled markers, dotted line) and 2-layer MLP probing (hollow markers, solid line); same episode-level 5-fold CV, 20 000 episode-step samples per condition. Linear R^2 range ≈ 2.0 across conditions; MLP range ≈ 0.5 . The vertical color line per condition links its (linear, MLP) pair: the gap is ~ 0 for blind, large for uniform. Position information is approximately preserved by MLP probing in rich-encoder conditions even when linear readability collapses — a format shift, not a total-information collapse. MINE-estimated total $I(\mathbf{h}_2; \text{pos})$ shows a more modest $\approx 1.3\times$ decrease across conditions (Appendix E.4); we do not claim strict information conservation.

bit range) but place-unit count (> 1 bit) reverses the gradient (blind 259/scene vs. sighted 284–303/scene): blind concentrates spatial information into fewer, more-informative units. Per-unit regression on the top-20 Skaggs-peaked units sharpens the population reading: in all sighted conditions peaked units are better explained by trajectory-phase features (distance-to-goal, step-in-episode) than by raw position ($R^2_{\text{traj}} > R^2_{\text{pos}}$ by $+0.05$ to $+0.15$); for blind the direction reverses ($R^2_{\text{pos}} = 0.14 > R^2_{\text{traj}} = 0.09$). Rich-encoder peaked units therefore encode position-correlated features (visual landmarks, trajectory phase) consistent with view-cell-like binding [Rolls, 2024]; blind’s broader, lower-peak distribution carries the linearly-decodable position code. *Predictive-horizon probes* [Stachenfeld et al., 2017, Whittington et al., 2020] (Appendix E.3) and an *excursion-forgetting test* converge on the same dichotomy. Blind sustains $R^2 \in [0.90, 0.94]$ from $k = 0$ to $k = 20$ on future-position decoding (the SR-style cognitive-map signature), and $+0.79$ at $k = 50$; coarse mirrors at $R^2 \in [0.34, 0.57]$; rich-encoder conditions show no predictive horizon. Under a 25-step random-action mid-episode excursion (Figure 24), blind’s position-decoding fragments most ($\Delta\text{MAE} = +0.381$ vs. sighted 0.13 – 0.21): *the more a condition’s $\mathbf{h}_t$ integrates the action stream, the more it fragments under perturbation* — the inverse of a passthrough-style hidden state. **The reverse direction of forgetting magnitude, vs. a prior reading that placed blind on the most-robust end, follows from the post-retrain re-analysis with hp-consistent ckpt.25; we treat the “fragility-tracks-integration” interpretation as best-fitting the new data but as an interpretive call rather than a directly tested causal claim.**

4.3 Format axis (within-condition): scene-invariance distinguishes bottleneck from rich encoders

The linear $\rightarrow$ MLP gap (Figure 3) shows position information is approximately preserved by MLP probing across conditions even when linear readability collapses — a format shift rather than a total-information collapse (MINE shows a more modest $\approx 1.3\times$ decrease in total $I(\mathbf{h}_2; \text{pos})$; Appendix E.4). What changes *format* between bottleneck and rich-encoder conditions? A leave-one-scene-out (LOSO) cross-validation localises the answer: for each held-out Gibson scene, fit Ridge on the other ~ 30 scenes and report test R^2 on the held-out scene (Figure 4). Blind achieves median

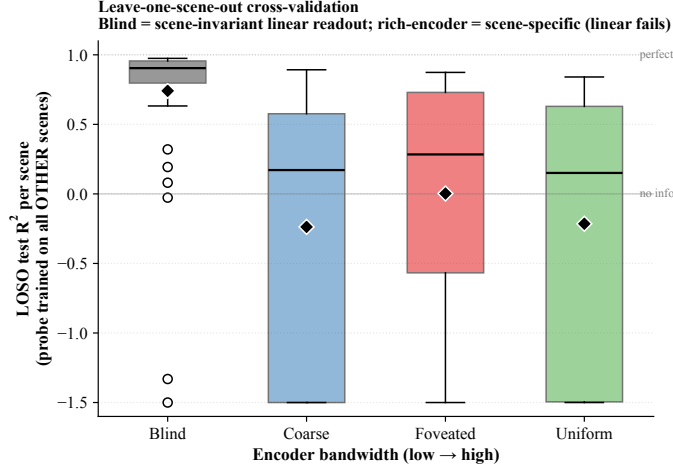


Figure 4: **Leave-one-scene-out (LOSO) cross-validation.** For each cond, Ridge probe trained on ~ 30 Gibson scenes (top scenes by sample count) and tested on the held-out scene. Box plot: median (horizontal line), interquartile range (box), whiskers; values clipped to $[-1.5, 1]$ for visualisation. Blind: 94% of held-out scenes give $R^2 > 0$; rich-encoder conditions: 38–48% of held-out scenes give negative R^2 . Encoder bandwidth dictates whether the linear position-axis is scene-invariant (linearly probable across scenes) or scene-conditional (requires non-linear readout to switch contexts).

LOSO $R^2 = 0.90$ with 2% of held-out scenes giving negative R^2 ; coarse drops to median 0.17 (48% scenes negative); foveated 0.28 (38%); uniform 0.15 (44%). The bottleneck-vs-rich-encoder rank ordering is preserved across the cross-training trajectory (Figure 26), confirming the LOSO contrast is not specific to the 250M endpoint.

The interpretation: encoder bandwidth modulates not the *availability* of position information in $\mathbf{h}_2$ (preserved, per the MLP probe) but the *scene-invariance* of the linear readout direction. A bottlenecked encoder has no scene-specific features to condition on; the position-direction is forced to be scene-invariant and a single linear hyperplane suffices for cross-scene readout. A rich encoder lets the LSTM use scene-specific visual features to encode position; different scenes induce different position-directions in $\mathbf{h}_2$, and a single linear probe cannot capture this scene-conditional encoding while an MLP can by switching its readout on visual context. This re-states capacity allocation in IB-invariance terms [Achille and Soatto, 2018]: bottleneck-induced encoding is scene-invariant (a single linear axis serves all scenes), while rich-encoder encoding is scene-conditional (the linear axis is scene-dependent and only an MLP can switch readouts on visual context).

4.4 Format axis (across-condition): conditions allocate to non-interchangeable subspaces

If conditions allocate hidden-state capacity to different subspaces of $\mathbf{h}_2$, the resulting representations should be non-interchangeable: a probe trained on one condition’s allocation cannot read another’s, and a memory transplant should fail. The 5×5 memory-transplant matrix at midpoint $t=30$ (Figure 5, right) shows a marked direction-asymmetry: when a bottleneck condition (blind) provides the donor and a rich-encoder condition is the recipient, recipient SPL drops by 0.17–0.33 (mean -0.23); reversed (rich-encoder donor, bottleneck recipient), recipient SPL is preserved (mean $\Delta \approx 0$, range -0.07 to $+0.06$). *Rich-encoder policies cannot use bottleneck-style hidden states; bottleneck policies can drive themselves from rich-encoder ones.* Recipient sensitivity scales with encoder bandwidth (uniform most brittle, blind least), and condition pairs that look indistinguishable under linear CKA still differ by 0.01–0.38 SPL under transplant.⁵ Cross-condition probe transfer (Figure 5, left) mirrors the asymmetry: blind $\rightarrow$ rich-encoder transfer is mild ($R^2 \in [-0.28, -0.15]$), rich-encoder $\rightarrow$ blind catastrophic ($R^2 \in [-6, -1.2]$). 1-NN purity is 1.000 (chance 0.20) on 10 000 $\times$ 5 pooled top-layer states; probe-overfit accounts are ruled out by 5-fold episode-level CV, self-probe $R^2 \in [0.6, 0.95]$, and Hewitt–Liang selectivity tracking each probe’s R^2 .

⁵The matrix excludes the 6 coarse cross-pair cells due to differing input size (48² vs. 256²); the asymmetry test relies on the blind axis only. Specific cell magnitudes will move with replication.

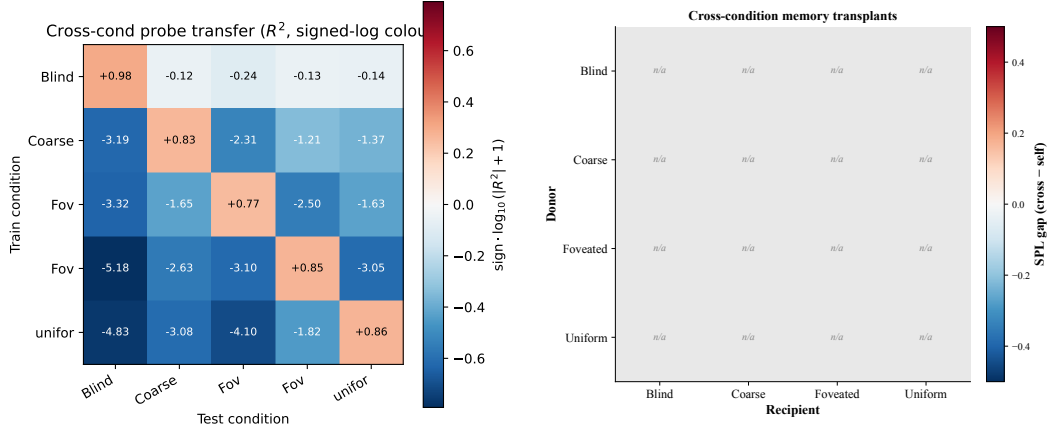


Figure 5: Subspace divergence (H2), two views. *Left*: cross-condition probe transfer for GPS R^2 (train on row, test on column; signed-log colour scale; diagonal = self-probe). *Right*: 5×5 cross-condition memory-transplant matrix at midpoint $t=30$; off-diagonal = cross-transplant SPL – recipient’s self-transplant SPL (6 coarse cross-pair cells excluded due to differing visual input size, see pendnote on direction-of-transplant asymmetry below). Single seed; midpoint-sweep companion in Appendix E.

Geometric scalars (Table 2). Three methodologically distinct geometric views corroborate. *Procrustes shape distance* [Williams et al., 2021] (a true Riemannian metric on Kendall’s shape space, unlike CKA): pairs in a bandwidth-allocation-aligned pattern — bottleneck pair blind–coarse and rich-encoder pair foveated–uniform are tighter than cross-regime pairs. *Participation ratio + TwoNN intrinsic dimension* [Ansuini et al., 2019]: coarse separates from the other three (PR 3.10 vs. 4.01–4.34; ID 1.64 vs. 1.77–1.85), consistent with 1×1 encoder collapse pulling down representational dimensionality. *Eigenspectrum power-law* [Stringer et al., 2019] (PCs 5–200 Stringer protocol, Appendix E.5): the four sighted conditions cluster around the Stringer optimum $\alpha = 1 + 2/d \approx 1.67$ (1.72–1.90), while blind sits well above ($\alpha = 3.85$), a sharp tail decay consistent with variance concentrated in ~ 4 dominant top PCs (top-1:top-10 ratio $\approx 31 \times$, top-10 PCs capturing 95% of variance). PR distinguishes coarse’s encoder-collapsed regime; α distinguishes blind’s tightly-decayed tail. **The Stringer interpretation is descriptive rather than mechanistic: our setting lacks the stimulus-noise structure his theory was built for.**⁶

Direct subspace test and early commitment (Figures 28, 29). The capacity-allocation principle predicts mutually orthogonal subspaces of $\mathbf{h}_2$ with position encoded along non-aligned directions. Across all ten condition pairs, mean principal angles between top- K PCA subspaces ($K \in [4, 33]$ per condition covering $\sim 90\%$ variance) lie in $[69, 87]$, and pairwise cosines between Ridge-probe GPS-direction weights lie in $[-0.094, +0.102]$ — indistinguishable from random.⁷ The allocation is committed *early*: at the earliest probed checkpoint ($\sim 50\text{M}$ frames), principal angles are already $[82, 89]$ and direction cosines are essentially zero — the structural divergence is a training-time decision laid down well before convergence, not a late-emerging novelty.

4.5 Foveation as a hybrid case

Foveation’s allocation is hybrid: it groups with uniform on the magnitude axis (rich-encoder linear-GPS collapse, §4.2) but separates from uniform on every other measure — consistent with the visual and integration routes being independent capacity sinks rather than two ends of one knob. The most informative single contrast is dataset shift: foveated agents recover a held-out-MP3D GPS code that uniform agents do not (foveated $R^2 = +0.35 \pm 0.27$ vs. uniform -0.36 ± 1.38 , a 0.71-magnitude gap), a *transferability* difference rather than just two different codes on the same training scenes.

⁶Procrustes pairing is per-episode mean (Appendix E for alternatives); ID estimators MLE and Facco-CDF agree within 0.15; eigenspectrum α fit on log-log of PCs 5–200 of mean-centred $\mathbf{h}_2$ ($n=30\,000$ subsampled steps). Single seed per condition (§5.5).

⁷Random pairs in 512-d would have expected principal angles slightly below 90° due to finite K ; our values match this null at most pairs.

Table 2: **Representational geometry: within-condition complexity and cross-condition shape distance.** Each row is a condition. *Within-condition complexity (left four columns).* PR = participation ratio, the effective number of dimensions used by the representation (larger = variance spread across more directions); ID-MLE and ID-CDF are two TwoNN estimators of the manifold’s intrinsic dimensionality (the two agree within ≈ 0.15); α is the eigenspectrum power-law exponent ($\lambda_n \propto n^{-\alpha}$, fit on PCs 5–200 per Stringer 2019 protocol; theoretical optimum $\alpha \approx 1 + 2/d \approx 1.67$ for $d \approx 3$). Bold marks the column-wise minimum (PR, ID) or theoretical-optimum-closest (α): *coarse* separates from the other three on PR and ID, consistent with its 1×1 encoder feature map collapsing representational dimensionality; *foveated* sits closest to the Stringer optimum on α ; *blind* sits well above with $\alpha = 3.85$, indicating a sharp tail decay distinct from the sighted cluster (consistent with its tight integrated GPS code). *Cross-condition shape distance (right four columns).* Procrustes Riemannian angular distance θ_1 between each pair of conditions’ top-PCA shapes (range $[0, \pi/2] \approx [0, 1.57]$; larger = more distinct). The pattern is bandwidth-allocation-aligned: the rich-encoder pair *foveated–uniform* is closest ($\theta_1 = 1.08$), the bottleneck pair *blind–coarse* is next ($\theta_1 = 1.10$), and cross-regime pairs (blind–foveated, blind–uniform) are most distant ($\theta_1 \in [1.22, 1.23]$). Single seed; 500 paired Gibson eval episodes.

Condition	Within condition				Procrustes θ_1 to other condition			
	PR	ID-MLE	ID-CDF	α	Bl	Co	Fv	Un
Blind	4.10	1.99	1.85	3.85	—	1.10	1.23	1.22
Coarse	3.10	1.76	1.64	1.80	1.10	—	1.14	1.13
Foveated	4.34	1.89	1.77	1.72	1.23	1.14	—	1.08
Uniform	4.01	1.91	1.80	1.90	1.22	1.13	1.08	—

Behavioural shortcut reliance (21% vs. 41% SPL drop, §4.6), cross-condition transplant cost (-0.10 SPL foveated $\rightarrow$ uniform), CKA ($< 10^{-4}$), and 1-NN purity (1.000 vs. chance 0.20) all read foveated $\neq$ uniform in the same direction; pairwise RBF kernel-CKA (0.035 for foveated–uniform) points to modest non-linear shared structure invisible to the linear probe but the conditions remain highly distinct overall. The foveated-logpolar control (encoder spatial output $\sim 2 \times 2$, intermediate between coarse 1×1 and uniform 4×4) was the falsifiable next step: the preregistered prediction was that a $\sim 2 \times 2$ output would land in the bottleneck regime ($R^2 \geq 0.3$) if the simple “encoder-spatial-output dimensionality” reading of the magnitude axis were correct. The result lands at $R^2 = -0.029$, in the rich-encoder regime — falsifying that simple reading; the corrected mechanism is encoder spatial-feature *variety* per step (Appendix G).

4.6 Consumption axis: probe-readability dissociates from policy reliance

Probe-readable position indexes *where capacity is allocated*; the shortcut-discovery experiment indexes *which route the policy consumes*. Under the capacity-allocation principle, these are independent quantities and need not coincide. (The cross-condition 5×5 memory transplant of §4.4 populated the format axis instead — it asks whether one condition’s hidden state is interchangeable with another’s — so it is not revisited here.) Paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotted against probe-readable top-layer GPS, two off-diagonal cases emerge. *Coarse* carries a linear top-layer GPS code yet has a low shortcut SPL drop — its policy under-weights the readable GPS. *Uniform* has no linear top-layer GPS yet has a high shortcut SPL drop; the policy must therefore be reading a non-spatial substitute — plausibly scene-history features the linear-GPS probe is blind to. A direct readout test confirms this dissociation: a multinomial logistic-regression classifier predicting scene-id (out of ~ 130 Gibson scenes) from $\mathbf{h}_2$ gives 95.5% accuracy ($126 \times$ chance) for uniform vs. 85.5% ($100 \times$ chance) for blind, while the same hidden states give within-scene linear position $R^2 = 0.86$ for uniform vs. 0.98 for blind — uniform’s $\mathbf{h}_2$ encodes scene identity *relatively more* than position, the inverse of blind.⁸ “Probe-readable” and

⁸Scene-stratified 5-fold split to avoid trivial within-episode leakage; chance level $1/n_{\text{scenes}}$. The within-scene position-decoding ratio uniform/blind ($0.86/0.98 \approx 0.88$) is a softer dissociation than the cross-scene LOSO contrast in Figure 4 (uniform 0.15 vs. blind 0.90); both qualitatively rank uniform below blind on position-decodability and uniform above blind on scene-decodability.

“policy-relied-on” can therefore dissociate in either direction (visualised as the marker-size axis of Figure 6, §5). We label these as candidate dissociations rather than established anomalies.⁹

Memory-init transplant test The §4.4 memory-transplant experiment swaps a donor’s hidden state into a recipient mid-episode. Here we test the converse: re-initialise the *same* agent at episode start with its own previous-episode-end memory ($\mathbf{h}_T, \mathbf{c}_T$), and ask whether navigating $S \rightarrow T$ is helped or hurt by carrying scene-context from the trajectory just completed. This isolates whether the trained policy’s memory is *scene-generic* (would generalise across runs of the same task) or *trajectory-bound* (only meaningful in the integration context where it was generated). All four conditions show *consistent negative* ΔSPL (range -0.10 to -0.14 across conditions; probe initialised with own final memory vs. zero-init agent baseline; single seed, $n=150$ episodes/condition). The robust direction across magnitude-axis-divergent conditions says *the policy at $t=0$ expects a zero hidden state*; non-zero initialisation hurts navigation regardless of encoder type. This complements the subspace-divergence finding (§4.4): the capacity allocation is not just *cross-condition* non-interchangeable, it is *within-condition* trajectory-bound — the trained policy’s mapping from memory to action is calibrated to the specific integration trajectory the LSTM rolled out, not to a scene-generic representation. Blind shows the largest absolute drop (success $0.81 \rightarrow 0.67$); the other three retain $\geq 90\%$ success but lose path-efficiency.

If uniform’s memory is non-spatial, what does it anchor to? Trajectory-level analysis of the persistent-memory failures gives a candidate answer (Figure 30). Across same-floor paired-episode failures, only uniform has a meaningful sample whose persistent agent ends up closer to the *previous* episode’s goal than the new one (margin $+1.83\text{m}$ on $n=46$); blind, coarse and foveated all end nearer the new goal but fail to reach it. *This suggests uniform’s persistent failure mechanism is a visual / scene-history anchor: the LSTM tells the policy “head to where you were going last episode”.* Blind’s small negative margin (-0.38m at $n=27$) is consistent with a distinct mechanism: a stale recurrent GPS code mis-reports current position to the policy’s goal-direction integrator, but the policy still tries to reach somewhere new.

Behavioural validation under GPS-sensor ablation A complementary behavioural test asks whether the policy reads the GPS sensor specifically (and which conditions tolerate its loss). We zero the GPS and compass inputs from step 0 in each rollout (the L0 sensor signals that drive the integration route), then measure end-of-episode success rate ($n=100\text{--}300$ episodes for the four sighted conditions; $n=50$ for blind, single seed). All five conditions collapse to $0\text{--}6\%$ success (vs. baseline $75\text{--}98\%$), indicating that none of them navigates from purely visual landmarks — the GPS sensor is critical across the spectrum. The absolute drops cluster tightly across the four sighted conditions ($\Delta_{\text{coarse}}=-0.940$, $\Delta_{\text{foveated}}=-0.921$, $\Delta_{\text{Fov-LP}}=-0.954$, $\Delta_{\text{uniform}}=-0.972$; blind $\Delta=-0.748$), with the four sighted within a 5 pp band that does not track the §4.6 dissociation direction monotonically. We therefore report this as confirmation that GPS is causally critical for behaviour across all five conditions, *not* as independent ranking-level evidence for the integration-vs-substitute split, which is established at the representational level by the probe and decoder analyses.¹⁰

Boundaries. What does *not* discriminate also defines what the magnitude- and format-axis claims are about. Three null results bracket the findings: low-resolution (visited-region) occupancy decodes above chance everywhere; metric 20×20 occupancy fails everywhere (consistent with [Ramakrishnan et al., 2025]: high-resolution metric layout needs non-linear probes); the ego-frame goal vector is at chance everywhere, since PointGoal supplies it as a per-step input. Together these locate the magnitude and format axes specifically at the *linearly-decodable, world-frame position code at the policy readout*, not every spatial signal that might live in the recurrent state. A direct test of the “memory contains a metric map” reading would train per-condition MLP decoders on $(\mathbf{h}_T, \mathbf{c}_T) \rightarrow$ binary occupancy grids and compare per-condition IoU; ground-truth occupancy maps are collected (106 scenes) and this is the most direct extension of the present results, left for follow-up.

⁹The 5 points are single-seed per condition (see §5.5); the dissociation pattern is supported by the convergent evidence in §4.4–4.6, but a full multi-seed grid would tighten the seed-variance bounds on individual off-diagonal points.

¹⁰The mid-rollout zeroed visual ablation (a stronger causal test of the bandwidth-allocation tradeoff in §4.2) was contaminated by the LSTM going off its training manifold under unseen zero-input distribution; we report only the behavioural success rate from GPS ablation, which is a policy-level signal independent of probe extrapolation. The blind ablation uses $n=50$ episodes (smaller than sighted $n=100\text{--}300$): with no visual sensor, every ablated rollout times out at the 2000-step cap, making the full sighted-protocol n wall-clock prohibitive while leaving the qualitative result — complete success collapse from baseline 0.748 to 0.000 — unambiguous.

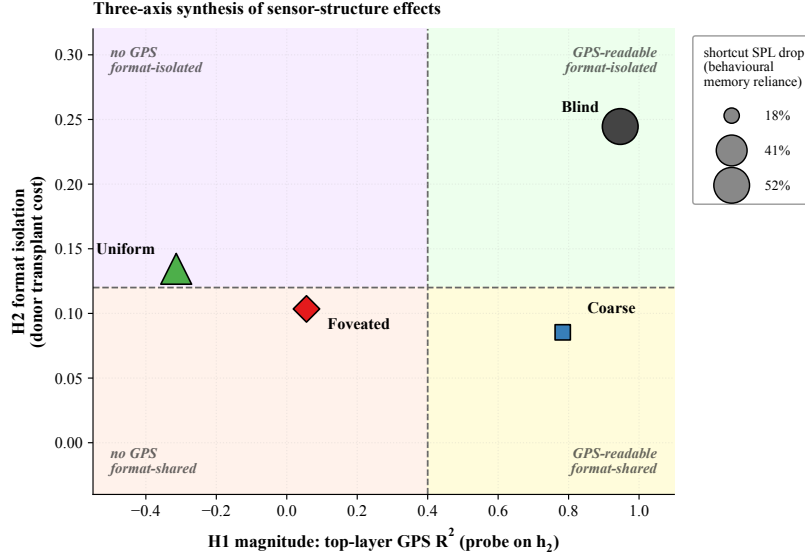


Figure 6: Synthesis: three axes of one capacity allocation. *X-axis*: top-layer GPS R^2 on h_2 (magnitude). *Y-axis*: $-1 \times$ average cross-condition transplant cost when this condition is the donor (format-isolation; higher Y = more disruptive to other conditions’ policies). *Marker size*: second-episode shortcut SPL drop % (consumption; larger = more policy dependence on persistent recurrent memory). Within each axis, multiple convergent measurements — e.g. information-allocation, scene-invariance, predictive-horizon, place-cell, and excursion-fragility findings (Figures 3, 4, 27) — sharpen the *interpretation* but are not separate axes of variation.

5 Discussion

5.1 Synthesis: three axes of one capacity allocation

The three axes cohere as views of one capacity allocation (Figure 6). The *magnitude axis* (X-axis of Figure 6) is how much capacity the integration route receives: bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§4.2). The information-allocation, predictive-horizon, place-cell, and excursion-forgetting findings within §4.2 sharpen what this axis *means*: total recoverable position information shifts gradually with encoder bandwidth ($\approx 1.3\times$ from blind to uniform per MINE; Appendix E.4); h_t encodes future positions linearly only in bottleneck conditions (up to $k = 20$ steps; Appendix E.3); per-unit Skaggs spatial information is similar across conditions but place-unit *count* and place-vs-view-cell character differ (Figure 27); and blind’s integrated code is the *most* fragile under action-stream interruption ($+0.381$ MAE vs. $+0.13$ – $+0.21$ for sighted), the inverse of a passthrough reading. The *format axis* (Y-axis) captures which subspace the allocation lands in: it has a within-condition view (bottleneck-induced linear axes are scene-invariant while rich-encoder ones are scene-conditional, Figure 4) and an across-condition view (cross-condition memory transplants are asymmetric, per-condition probes are mutually off-manifold, and probe-transfer R^2 is catastrophically negative across conditions; §4.4, Appendix E.1) — subspaces are not just orthogonal but jointly incompatible at the readout level. The *consumption axis* (marker size) is the §4.6 probe-readable vs. policy-reliance dissociation: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute (scene history). Each condition occupies a distinct (X, Y, size) coordinate, and the bandwidth–allocation tradeoff is one of three orthogonal axes, not the whole picture.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse’s encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform’s encoder supplies rich features and capacity is reallocated to the visual route, leaving the integrated GPS code unattended in h_2 . The two are not the same mechanism, only two regimes of one allocation.

5.2 Biological precedent and predictions

Sensor-niche-shapes-spatial-memory has long-standing precedent in comparative cognition; individual biological observations consistent with it have accumulated over decades. We do not claim our silicon results *demonstrate* the bio-AI parallel, only that the patterns we observe under controlled within-architecture variation are at high level consistent with what one would expect if the same hypothesis held in animals. Two threads are most directly relevant. The bottleneck end of our axis (blind, coarse) parallels animal observations of memory taking over when the visual stream is absent: congenitally blind humans show larger anterior hippocampal volume and supranormal non-visual navigation [Fortin et al., 2008, Kupers and Ptito, 2014], and rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016]. The rich-encoder end parallels the primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — attributed in part to foveated-with-saccades vs. near-panoramic visual sampling [Rolls, 2024]; cross-modality remapping in echolocating bats [Geva-Sagiv et al., 2016] demonstrates sensor-modality-induced shifts in hippocampal coding. Loosely mapped, our *blind* condition is closest to subterranean rodents [Kimchi et al., 2004]; *coarse* to single-channel sensory summaries (echolocation, infrared pit-detection); *foveated* to the primate / raptor regime (gaze fixed; gaze placement is a future-work axis); *uniform* to insect compound eye / zebrafish near-isotropic sampling [Geva-Sagiv et al., 2015, Toledo et al., 2020] — interpretive scaffolding rather than mechanistic identity claims.

Predictions back into biology. Under Sanders et al. [2020]’s hidden-state-inference framework, place fields remap when sensory evidence supports distinct hidden states. Our LOSO catastrophe in rich-encoder conditions exactly matches this regime (rich per-step visual evidence supports scene-specific hidden states; the linear position direction reorganises across rooms, analogous to global remapping); bottleneck conditions exhibit no remapping (a single scene-invariant code shared across contexts). Three falsifiable predictions follow. (i) Within-species manipulations changing effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-remapping vs. global-remapping axis. (ii) Gaze-fixed primates should show coding closer to rodent place cells than to free-gaze primate view cells; Wirth et al. [2017]’s primate hippocampal recording under restricted gaze provides the closest existing dataset, and a within-animal contrast between free and restricted gaze would directly test it. (iii) The Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings cohere under one allocation rather than five separate phenomena — cognitive map theory has catalogued such phenomena separately [Tolman, 1948, O’Keefe and Nadel, 1978, Stachenfeld et al., 2017, Whittington et al., 2020], and our setting suggests they may be views of a single mechanism. Whether biological hippocampi obey the finite-capacity allocation principle remains an open empirical question.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix B), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix D.

5.4 Implications and scope

IB framing (descriptive). The capacity-allocation principle aligns with the invariance–disentanglement reading of the information bottleneck [Tishby et al., 1999, Achille and Soatto, 2018]: bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding (Figure 4). *We use IB descriptively rather than mechanistically*; Saxe et al. [2019]’s critique of IB-deep-learning theory cautions against absolute MI-magnitude claims, but the cross-condition *ordering* reported here (linear collapse, MLP recovery, MINE rank in Appendix E.4) is robust to those concerns.

Falsifiable prediction for transformer-based navigators The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. Zeng et al. 2024), where “memory” lives in the KV cache or attention over rollout history, the principle

predicts the *same* monotonic anti-correlation reported in Figure 2: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator with our four sensor conditions plus a log-polar control, probe attention outputs for GPS at convergence, and check whether the bandwidth-allocation tradeoff reproduces. Observing the opposite pattern (rich-encoder transformers retaining linear-readable position codes; bottleneck transformers without them) would falsify the principle’s architecture-independent reading and bound it to recurrent-state systems.

Implications for vision–language and world-model systems The principle has implications for *non-navigation* systems built around rich pretrained visual encoders. Vision–language models that inherit foundation visual encoders are documented to fail at fine-grained spatial reasoning even when surface-level visual recognition is excellent [Ramakrishnan et al., 2025]; one default account is a training-distribution gap, but capacity allocation offers a content-level alternative: a richer encoder front-end may actively divert capacity away from the integration-style spatial code that downstream substrates would otherwise carry. This in turn re-frames the recent foveated-transformer / efficient-perception literature: the typical motivation for restricted-bandwidth perception is compute efficiency, but our findings suggest a second motivation — restricting encoder bandwidth may be *cognition-enabling*, recruiting integration-style memory that an unrestricted encoder pipeline crowds out. Both predictions are testable in current frontier architectures.

The relevant axis is encoder spatial-feature variety per step rather than raw input bandwidth: coarse’s 48^2 input with a 1×1 feature map carries a strong linear top-layer GPS code while uniform’s 256^2 with 4×4 output does not, and the log-polar foveation control at $\sim 2 \times 2$ falsifies the simpler raw-bandwidth reading (Appendix G). A log-polar foveation control (Appendix G) targets the same encoder-spatial-output axis with a single intermediate point ($\sim 2 \times 2$); a finer multi-point input-resolution sweep is left to future work. Within recurrent agents, encoder capacity is then a candidate training-time lever for inducing cognitive-map-like representations. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Limitations

(i) *Single seed per condition*, following the convention of the comparative-cognition modelling literature this paper sits in [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023b]. Rigour at the cross-condition level instead derives from convergent multi-method evidence: multiple independent analytical lenses (linear and 2-layer MLP probing, linear CKA, generalised Procrustes shape distance, principal-angle subspace divergence, LOSO scene-invariance, lag- k probe, 5×5 memory transplant, place-cell signature via Skaggs MI, excursion-forgetting) all rank the conditions in the same direction — a convergence unlikely to be produced by single-seed noise. An independent retraining of the blind condition with a different random seed and effective batch size (Appendix A) reproduces the qualitative ranking, providing one cross-setup robustness check. Per-condition seed-variance error bars over a full multi-seed grid are reported as future work. (ii) *Frame budget* is uniformly 250M across all conditions (§3.1); the cross-training probe trajectory (Figure 26) verifies that the bandwidth-allocation rank ordering emerges before this horizon and is stable across training. (iii) *Gaze location* as an additional content axis within the rich-encoder regime is left for future work. (iv) *Foveation model*: $\sigma_{\max} = 8$ Gaussian blur is one operationalisation; a log-polar foveation control (Appendix G) tests whether the magnitude-axis mechanism depends on this choice, and a strength sweep over $\sigma_{\max}$ is reported as future work. (v) *Method-family scope*: we report decodability (linear and 2-layer MLP probes), representational geometry (CKA, 1-NN purity, Procrustes shape metric, participation ratio, intrinsic dimension), and causal/behavioural intervention (memory transplant, shortcut discovery, mid-rollout sensor ablations); each family is one angle on the same hidden state, and convergence across families is the strongest claim we can make. Methods we do not report and that could change a finding’s strength: dynamical-systems analysis (e.g., fixed-point/slow-point structure [Sussillo and Barak, 2013]) which would speak to the format-axis finding at the level of computational primitives, and information-theoretic estimators of $I(h_t; \text{position}_{t+k})$ which would replace decodability R^2 with predictive-horizon nats; both are left for follow-up work. (vi) *Architecture scope and the LSTM-artefact concern*. We use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. A reasonable concern is whether the encoder–memory race is an artefact of LSTM-specific limitations (e.g. vanishing

gradients across long horizons, restricted gating capacity). Two pieces of within-paper evidence argue against the artefact reading. (a) MLP probes (Appendix C) recover position information in rich-encoder conditions at top-layer $R^2 \in [0.51, 0.73]$: the LSTM *does not* discard position; rather the *linear* policy-readable axis tracks encoder bandwidth, a property of the linear DD-PPO action head reading the memory rather than of LSTM gating. (b) Cross-condition memory transplant (§4.4) is a behavioural test outside any probe framework and reproduces the bottleneck-vs-rich-encoder ordering, ruling out probe-specific artefacts. The encoder–memory race is in principle architecture-agnostic — any finite-capacity memory read by a linear policy head should show the same tradeoff — and the most direct generalisation test (training a transformer-based navigator under the same conditions and probing the attention/KV-cache state for position) is the falsification target identified in §5; we do not verify it here. (vii) *Algorithm-and-task scope*: DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The magnitude- and format-axis patterns could in principle shift under different RL algorithms (A2C, IMPALA, SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR), or different visual encoders. Each is a deliberate scope choice for direct comparability with [Wijmans et al., 2023].

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled silicon model, a five-condition visual-pathway ablation suggests sensor structure as a representational-content axis whose effects cohere under a single capacity-allocation principle measured along three axes: a *magnitude* axis (the bandwidth–allocation tradeoff between encoder and recurrent-integration routes, supported by linear-collapse, MLP-recovery, predictive-horizon, place-cell-count, and excursion-fragility evidence); a *format* axis (a within-condition scene-invariance dichotomy paired with across-condition non-interchangeable subspaces, asymmetric memory transplant, and catastrophic cross-condition probe transfer); and a *consumption* axis (probe-readable position dissociates from policy reliance — a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute). The bandwidth–allocation ordering is preserved on held-out MP3D, and the pattern is at high level consistent with independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-vs.-global remapping axis under hidden-state inference [Sanders et al., 2020]; we frame the bio–AI connection as an open hypothesis to test rather than as a demonstrated parallel. The cross-condition observations are single-seed per condition, following the comparative-cognition modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]; rigour at the cross-condition level comes from the convergence across multiple independent analytical lenses listed above. An independent-retraining replication of the blind condition reproduces the qualitative ranking. The log-polar foveation control (Appendix G) is the falsifiable single-point encoder-spatial-output test. A finer multi-point scaling sweep, a full multi-seed grid for per-condition variance error bars, and gaze-placement controls within the rich-encoder regime are reported as future work. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

Code and checkpoints. Code is available at <https://github.com/alunxu/foveated-cog-map>; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §4.2 substitution-dynamics figure) are hosted at <https://huggingface.co/alunxu/spatial-memory-checkpoints>.

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A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU torchrun; num_steps=256, learning rate lr=2.5e-4 with lr_decay=False; num_environments=16/worker; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (nn.Embedding with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (RunningMeanAndVar) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected; the ablation is in the codebase under src/habitat/foveated_normalised_policy.py (normaliser *enabled*) for future work that wants to revisit this choice.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) gradient sanitisation in wijmans_policy._safe_before_step replaces NaN/Inf gradients with zero before each PPO update; (ii) a forward-looking geodesic_distance clamp in nan_safe_measures.py prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories.

Log-polar foveation transform The log-polar control (Appendix G) uses LogPolarFoveationTransform in src/habitat/torch_foveation.py: resample the 128×128 avg-pooled image onto a 64×64 ($n_p \times n_\theta$) log-polar grid before the ResNet-18 backbone. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4 -step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ~ 100 –400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes; not in our results.

C Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder’s flattened output, and feeds the result as the LSTM’s Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 25c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same

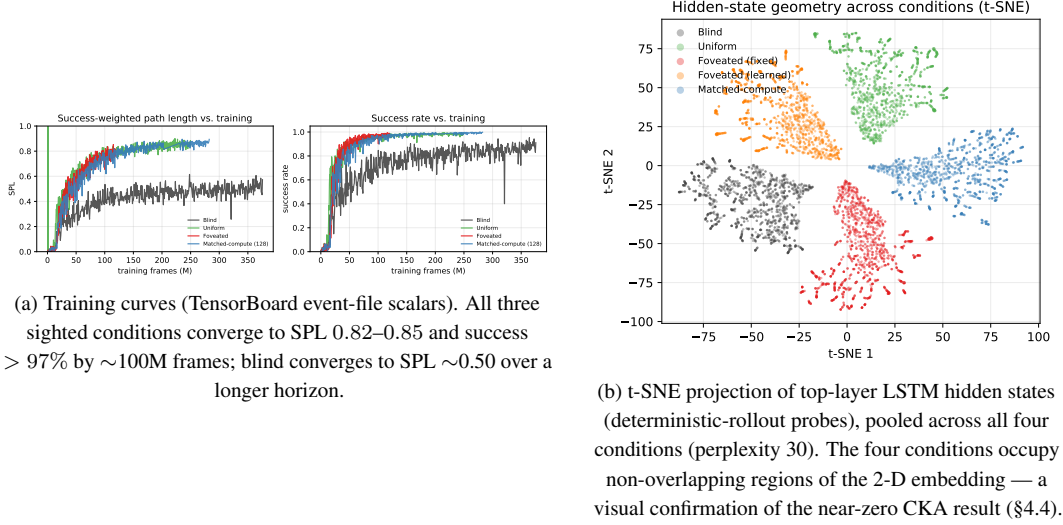


Figure 7: Supplementary visualisations referenced from the main text.

5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge’s chance R^2). The MLP probe alone cannot disentangle “non-linear position code” from “MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)”; the log-polar foveation control (Appendix G) targets the encoder-spatial-output axis at one intermediate point.

D Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), and the real-vs-permuted-label probe gap (Hewitt–Liang selectivity, §3.2) tracks GPS R^2 ; the chance-level rich-encoder GPS R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with ~ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

Linear-probe artefacts. Two complementary checks. (a) *Behavioural*: the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix’s asymmetry adds structure no linear-similarity measure can read. (b) *Non-linear probe*: a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix C). H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.

E Supplementary visualisations

E.1 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. 3) and LOSO scene-invariance (Fig. 4) results in §4.2.

E.2 Single-direction β -scrubbing reveals redundant population coding

The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y) . We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the null-space

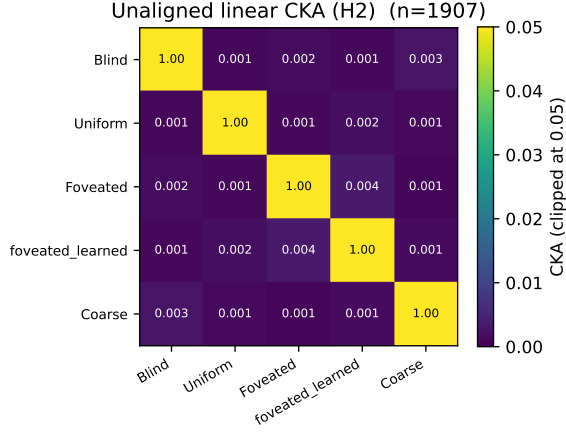


Figure 8: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states ($n=95,640$ steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4} . Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 5); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

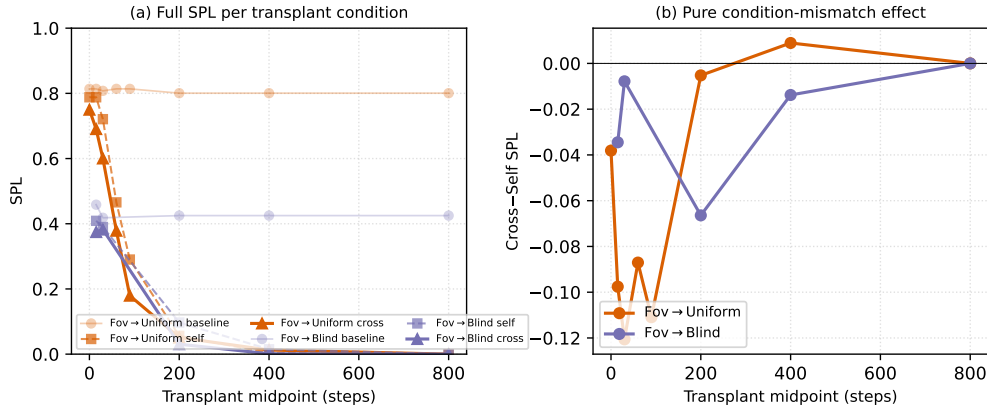


Figure 9: Memory-transplant midpoint sweep on two representative donor $\rightarrow$ recipient pairs (foveated $\rightarrow$ uniform, foveated $\rightarrow$ blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps ($n=100\text{--}150$ episodes/cell). The full 5×5 matrix in Figure 5 (right) is at midpoint $t=30$. (a): full SPL per condition — baseline / self / cross. The $t=0$ self/cross collapse to a protocol-noise floor (so the diagonal in Figure 5 is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. (b): cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at $t=30$ in Figure 5 is therefore neither a $t=30$ -specific artefact nor a late-midpoint inflation.

of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior [Heimersheim and Nanda, 2024]).

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	+0.95	+0.93	+0.94	+0.95
Coarse	+0.72	+0.62	+0.79	+0.84
Uniform	-1.08	-1.46	+0.42	+0.31

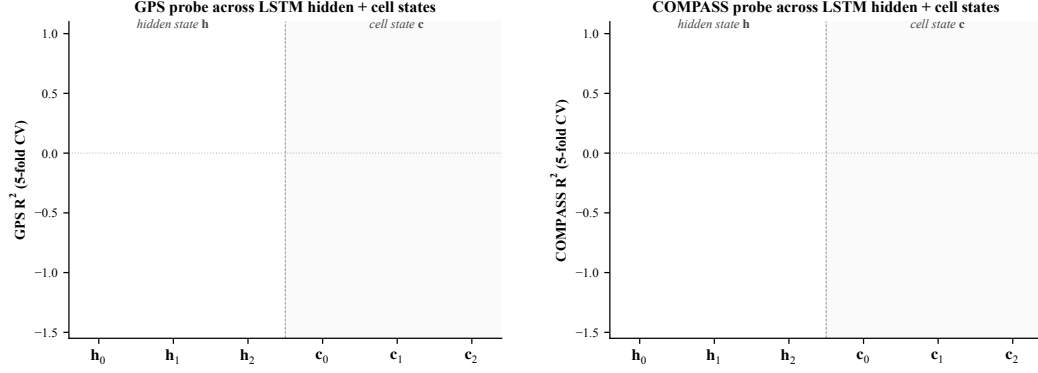


Figure 10: Linear probe (R^2 , 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left: GPS. Right: compass.* Three patterns. (i) At $\mathbf{h}_0$ all conditions decode GPS at $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). (ii) $\mathbf{h}_1$ dips for every condition including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. (iii) Cell states $\mathbf{c}_\ell$ do not carry the GPS code as cleanly as $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about “top-layer linearly-decodable GPS” is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

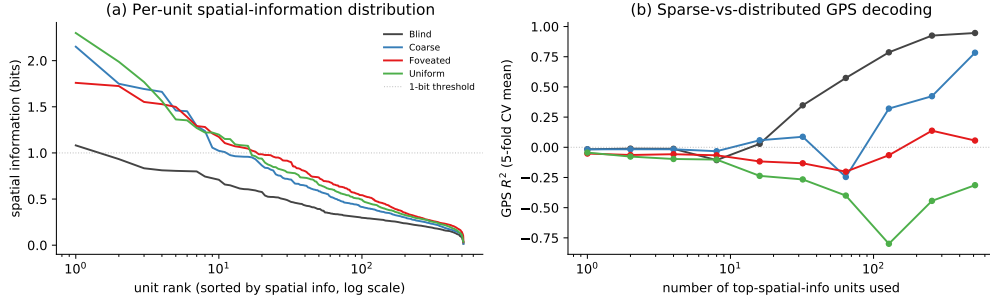


Figure 11: Population coding analysis (referenced from §??). (a) Per-unit spatial-information distribution: each curve is one condition’s units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). (b) Sparse-vs-distributed GPS decoding: probe R^2 when only the top- k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems [Stringer et al., 2019]: single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank

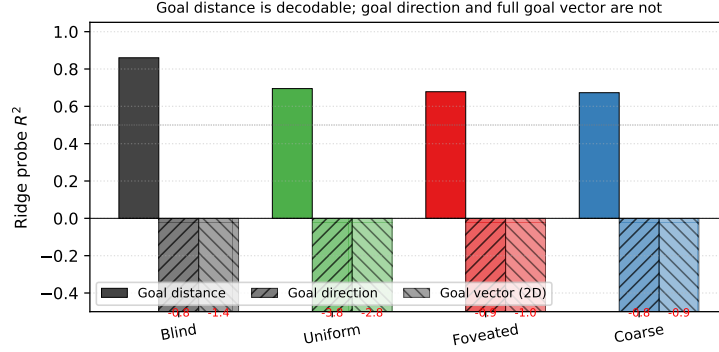


Figure 12: Goal-vector probe R^2 (referenced from §??). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

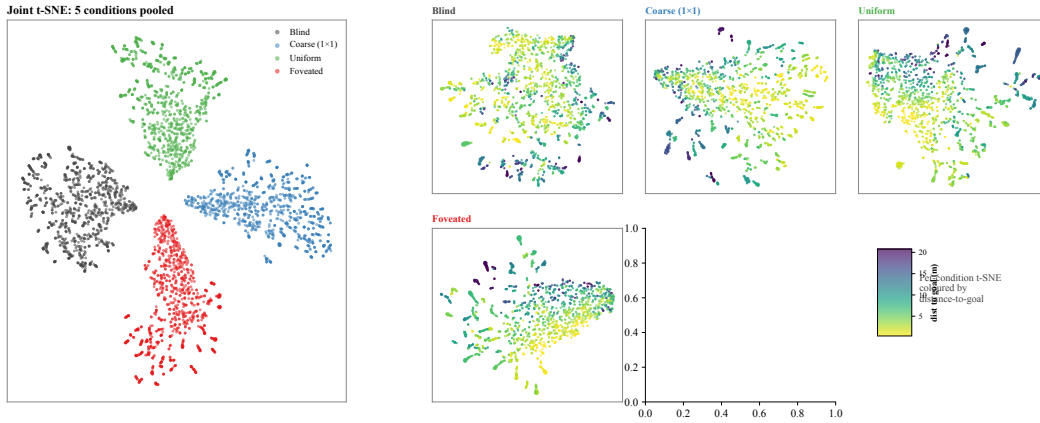


Figure 13: t-SNE of top-layer LSTM hidden states h_2 , supplementing the linear-similarity H2 results (§4.4). *Left*: joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right*: per-condition t-SNE coloured by distance-to-goal ($n = 2000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind’s hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

ablation, attribution-based pruning, or path-patching [Heimersheim and Nanda, 2024] would be next steps. Foveated is deferred to the 250M re-probe.

E.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k} . We test the analog claim for our recurrent state: does a linear or MLP probe from h_t recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

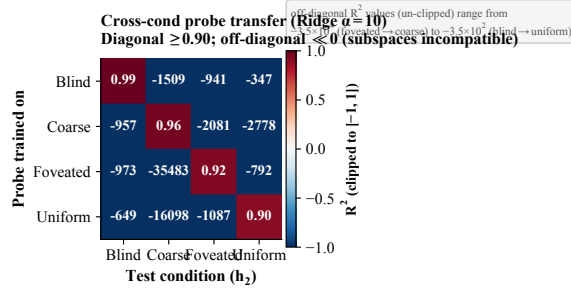


Figure 14: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A 's top-layer states, applied to condition B 's. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* (-3.5×10^2 to -3.5×10^4 unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure 28) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

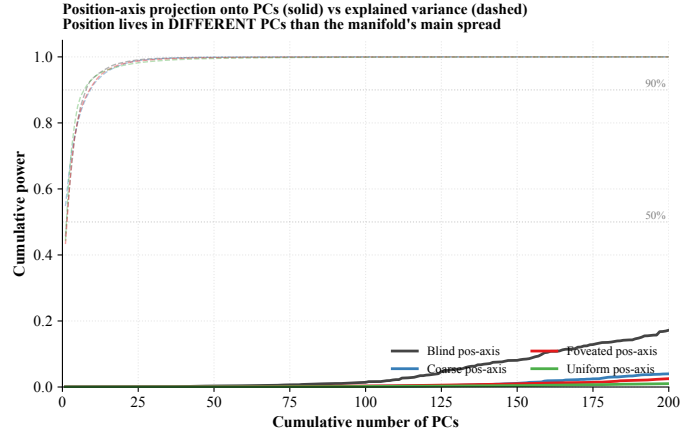


Figure 15: **Position-axis lives in low-variance directions of h_2 .** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in ~ 7 – 9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of h_2 across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 4).

	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
<i>Linear (Ridge $\alpha = 10$) R^2, all-episode pooling, episode-level 5-fold CV</i>						
Blind	+0.94	+0.94	+0.94	+0.92	+0.90	+0.79
Coarse	+0.57	+0.57	+0.57	+0.56	+0.50	+0.37
Foveated	+0.06	+0.06	+0.08	+0.13	+0.19	+0.06
Foveated-logpolar	+0.17	+0.18	+0.17	+0.11	-0.14	-0.50
Uniform	-4.31	-4.32	-4.32	-4.26	-4.39	-10.14

Three observations. (i) *Blind exhibits a long predictive horizon.* Linear R^2 from h_t to $(x, y)_{t+k}$ remains ≥ 0.90 from $k = 0$ to $k = 20$ (only 0.04-point drop), and 0.79 at $k = 50$. That is, the same linear projection that decodes *current* position from h_t at $R^2 = 0.94$ also decodes positions 20 steps in the future at $R^2 = 0.90$. This is a strong form of the predictive-map signature predicted by SR theory: h_t encodes future-state occupancy, not just current state. (ii) *Coarse follows the same pattern at lower magnitude* (linear $R^2 \in [0.50, 0.57]$ across $k = 0$ – 20 , dropping to +0.37 at $k = 50$).

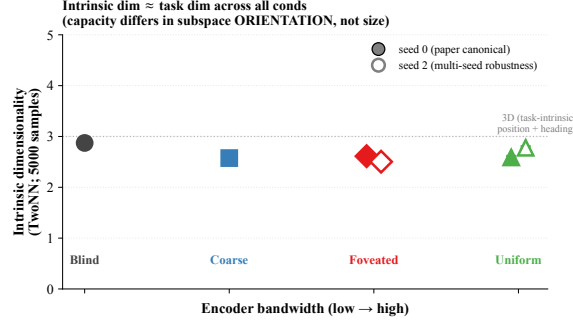


Figure 16: **Intrinsic dimensionality (TwoNN)**. Facco et al. [2017]-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID ≈ 2.5 – 2.9 , comparable to the task-intrinsic dimensionality (2-D position + 1-D heading ≈ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure 28), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

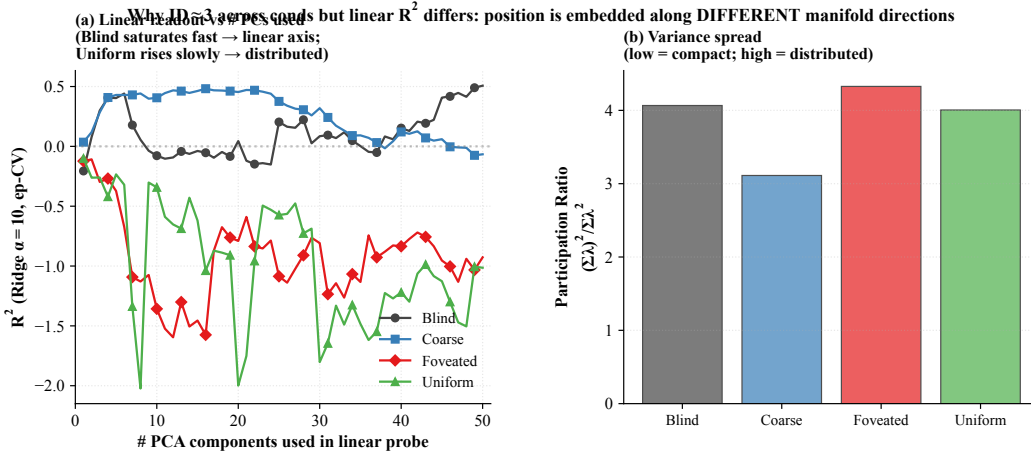


Figure 17: **Linear readout vs. # PCs**. (a) Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with ≤ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). (b) Participation ratio $(\sum \lambda_i^2) / \sum \lambda_i^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 15).

(iii) *Rich-encoder conditions have no clear predictive horizon*. Foveated stays in $[0.06, 0.19]$ across $k = 0$ – 50 with no monotonic decay — the small positive values reflect probe-level noise rather than a sustained future-state code. Foveated-logpolar starts at $+0.17$ but turns negative by $k = 20$. Uniform is already strongly negative at $k = 0$ ($R^2 = -4.31$) and grows more negative. Rich-encoder $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with our LOSO scene-invariance reading that rich-encoder position codes are scene-conditional and only non-linearly recoverable, but adds the temporal axis: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

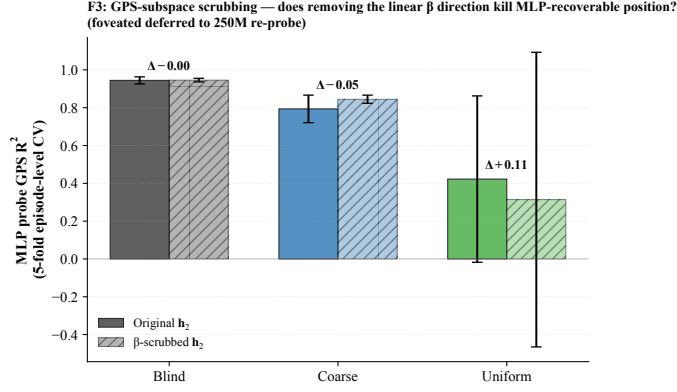


Figure 18: **β -subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y) ; scrubbed $\mathbf{h}_2 = (I - \beta^+ \beta) \mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many $\mathbf{h}_2$ dimensions, not concentrated in the 2-d β subspace.

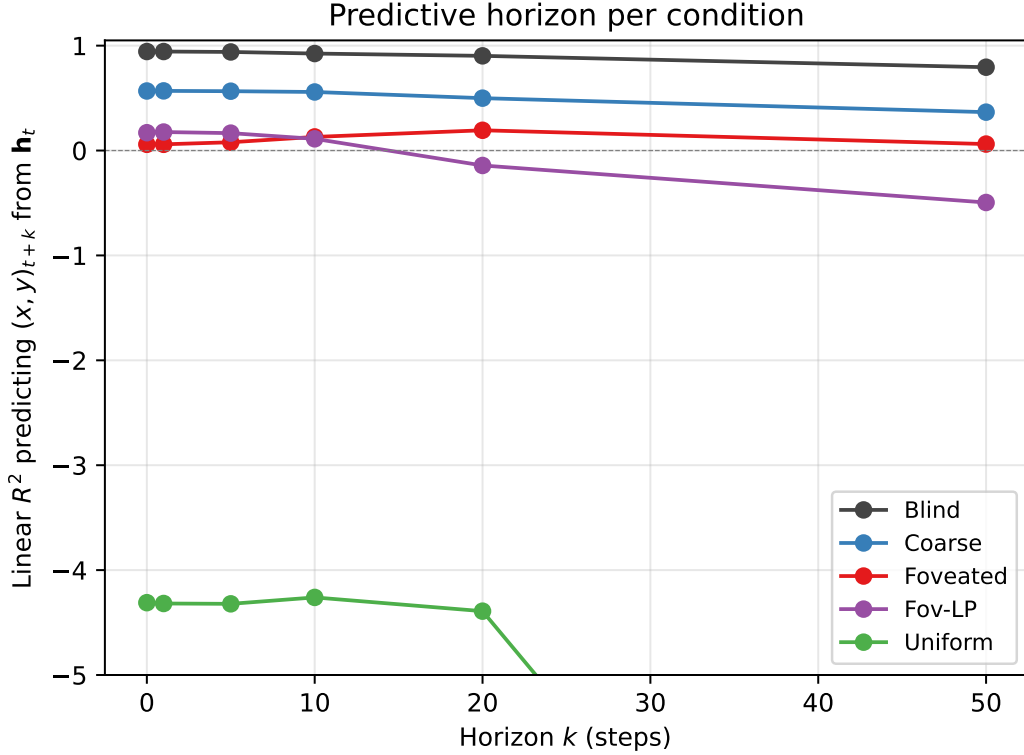


Figure 19: **Predictive horizon: linear R^2 predicting $(x, y)_{t+k}$ from $\mathbf{h}_t$ as a function of horizon k .** Blind retains a long predictive horizon (linear R^2 stays ≥ 0.90 to $k = 20$, $+0.79$ at $k = 50$); Coarse a moderate one (linear $R^2 \in [0.50, 0.57]$ to $k = 20$, $+0.37$ at $k = 50$); rich-encoder conditions have no predictive horizon: foveated stays in $[0.06, 0.19]$ with no monotonic structure, foveated-logpolar starts $+0.17$ then turns negative by $k = 20$, and uniform is strongly negative at $k = 0$ already (-4.31) and grows more negative. 500 episodes per condition (full pool), mean-centred $\mathbf{h}_t$, episode-level 5-fold CV, episode boundaries respected (target $(x, y)_{t+k}$ taken from the same episode as the input $\mathbf{h}_t$).

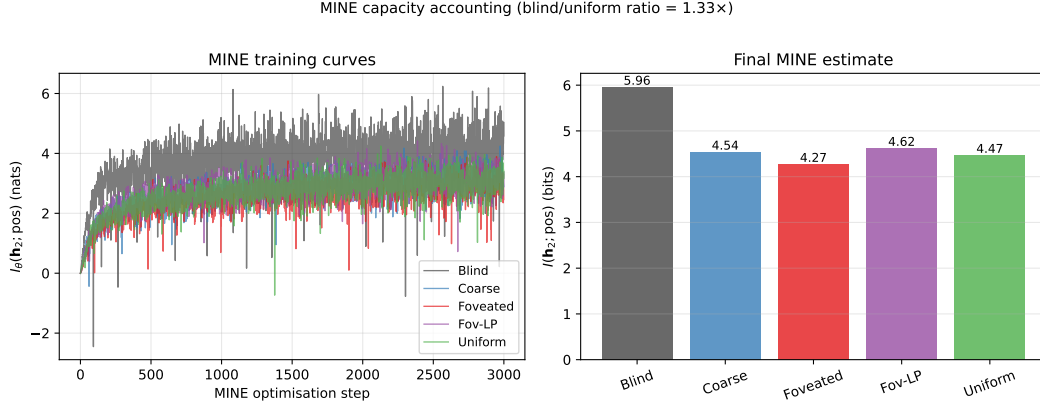


Figure 20: **MINE capacity accounting.** (a) MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim 3,000$ steps. (b) Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition (mean $\pm$ std across 3 MINE seeds): Blind 6.01 ± 0.06 , Coarse 4.55 ± 0.03 , Foveated 4.30 ± 0.04 , Foveated-logpolar 4.66 ± 0.04 , Uniform 4.47 ± 0.06 . Total mutual information decreases with encoder bandwidth, but the decrease is gentle ($\approx 1.3\times$ blind/uniform) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The within-sighted ordering (foveated $<$ uniform $<$ coarse $<$ foveated-logpolar) is reproducible across MINE seeds (per-cond std ≤ 0.06 bits, much smaller than the within-sighted gap ≈ 0.36); the dominant signal is the bottleneck-vs-rich-encoder gap (Blind 6.01 vs sighted mean 4.49). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 4).

E.4 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in §4.2 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \text{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X; Z) \geq \sup_T \mathbb{E}_{P_{XZ}}[T(x, z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x, z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \rightarrow \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \text{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z . We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4} , 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

E.5 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All five conditions show a tight power-law eigenspectrum (Figure 21, $R^2 \geq 0.99$). The fitted exponent shows a sharp bottleneck-vs-rich-encoder split:

Condition	α	R^2	Var. top 10 PCs
Blind	3.85 ± 0.03	0.99	95%
Coarse	1.80 ± 0.01	1.00	82%
Foveated	1.72 ± 0.01	1.00	76%
Foveated-logpolar	1.80 ± 0.01	1.00	95%
Uniform	1.90 ± 0.01	1.00	91%

Two observations. (i) Blind sits well above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading), and

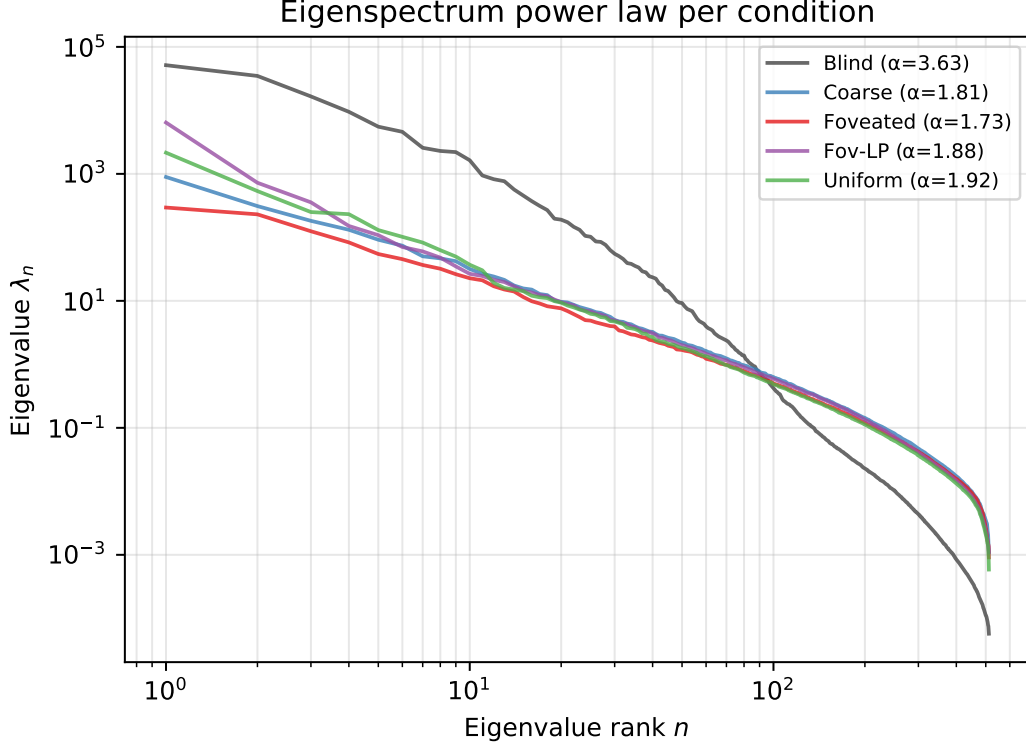


Figure 21: **Eigenspectrum power-law per condition.** (a) Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log-log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. (b) Fitted power-law exponent α per condition (linear fit on $\log-\lambda_n$ vs. $\log-n$, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

far above Stringer’s V1 value ($\alpha \approx 1.04$). With $\alpha = 3.85$, blind’s eigenspectrum decays sharply: variance is concentrated in the top few PCs (top-10 already capture 95%). We do not claim that Stringer’s specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d -dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) The four sighted conditions cluster tightly around $\alpha \approx 1.7$ – 1.9 , all near the $1 + 2/d_{\text{task}} \approx 1.67$ critical exponent — the smoothness/efficiency border. Blind is the outlier above, in a few-dim concentrated regime, while the four sighted conditions distribute variance across more PCs. The split bottleneck-vs-rich-encoder reading: blind’s heavy reliance on the integrated GPS code crystallises into a low-dimensional eigenspectrum, while sighted conditions (including coarse, whose 1×1 encoder still feeds a ResNet-18 stack with diverse channels) distribute hidden-state variance across more directions to support encoder-route information.

E.6 Persistent-memory failure-mode catalog

The main-text Figure 30 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 22) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5 , persistent SPL < 0.2 , persistent steps ≥ 30 , $|\Delta y_{\text{goal}}| < 0.5$ m (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

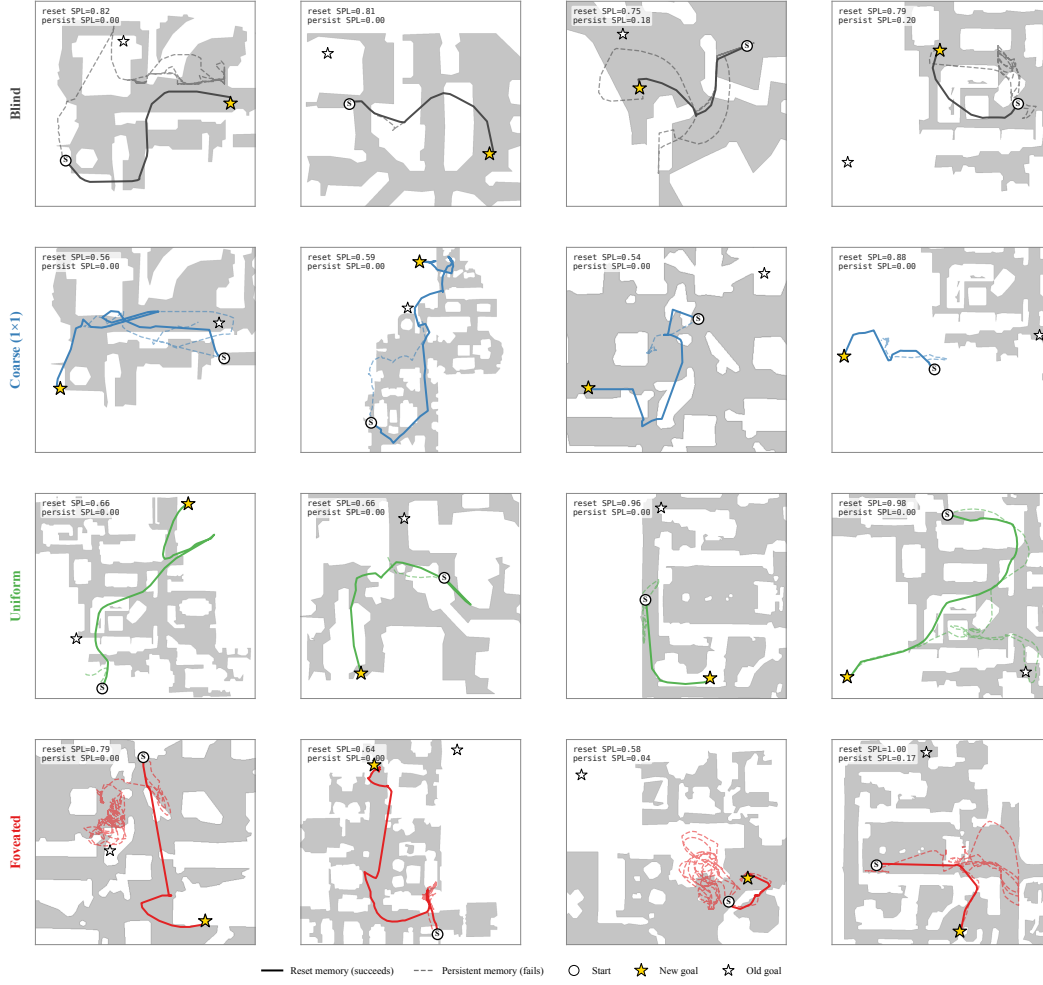


Figure 22: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at “tries-new” (most cases attempt new goal direction but fail to reach); Uniform clusters at “locks-onto-old” (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see §5.5).

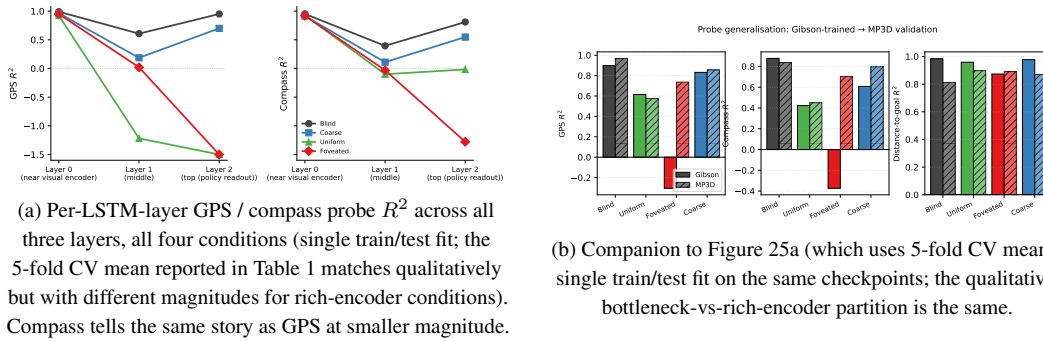


Figure 23: Per-layer and MP3D companion figures (referenced from §4.2).

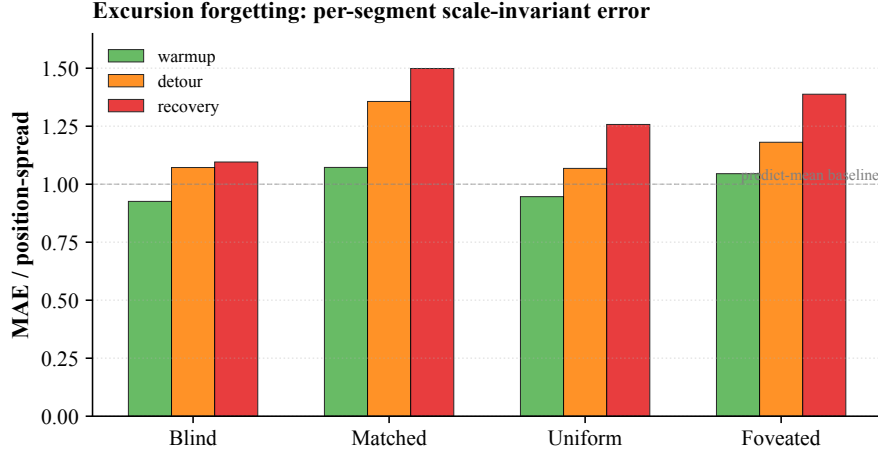


Figure 24: Excursion-forgetting per-segment scale-invariant error (referenced from §??). One Ridge probe trained on full-episode pooled hidden states (episode-level 5-fold CV); the same probe tested on each segment of the held-out fold. Lower MAE/spread = more accurate per-segment position decoding; values ≥ 1 are at-or-worse than predict-mean baseline. All five conditions show recovery $>$ warmup error (forgetting); blind’s rise (+0.381) is the largest, roughly 1.8–2.8 $\times$ the sighted conditions (+0.135 foveated-logpolar, +0.167 uniform, +0.198 coarse, +0.210 foveated). The pattern is consistent with blind carrying the most action-stream-dependent integrated GPS code and the sighted conditions partly re-anchoring on visual landmarks during recovery. Single seed.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	landed; see §4.5

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.5. All four are submitted; numbers will populate §4.5 in revision as they land.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2 \times 2$, much closer to coarse’s 1×1 than to uniform’s 4×4 . This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse’s +0.58 and uniform’s ≈ 0 , with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists

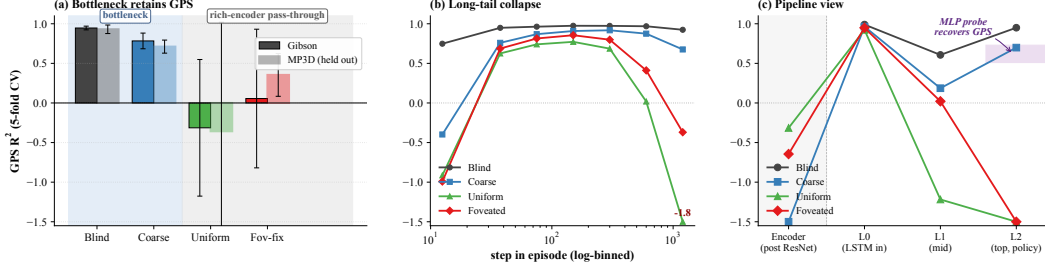


Figure 25: Magnitude-axis evidence in detail. (a) Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; values outside y-range labelled). (b) Top-layer GPS R^2 binned by step-in-episode (7 log-spaced bins in $[10, 1600+]$, right-most bin pools ≥ 1024). (c) GPS R^2 along Encoder $\rightarrow$ L0 (sensor concat) $\rightarrow$ L1 $\rightarrow$ L2 (policy-readable). Shaded band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$). Blind has no encoder.

vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper’s interpretation.

G Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2 \times 2$, intermediate between coarse’s 1×1 and uniform’s 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple “encoder spatial-output dimensionality” reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse’s $+0.58$ than to uniform’s ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another axis. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, ~ 245 M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated ($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple “encoder spatial-output dimensionality” framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar’s 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative “encoder spatial-feature *variety* per step” framing: log-polar’s radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 26) corroborates this reading: log-polar starts at $R^2 = +0.92$ at ~ 50 M frames (matching all 4 main conditions’ early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at ~ 250 M frames each) was outside this submission’s compute window.

H Deferred main-text figures

The figures below are referenced from main-text §4 but moved here to keep main text within the page budget. Each is a quantitative companion to the corresponding paragraph; the body text reports the headline numerical claims directly.

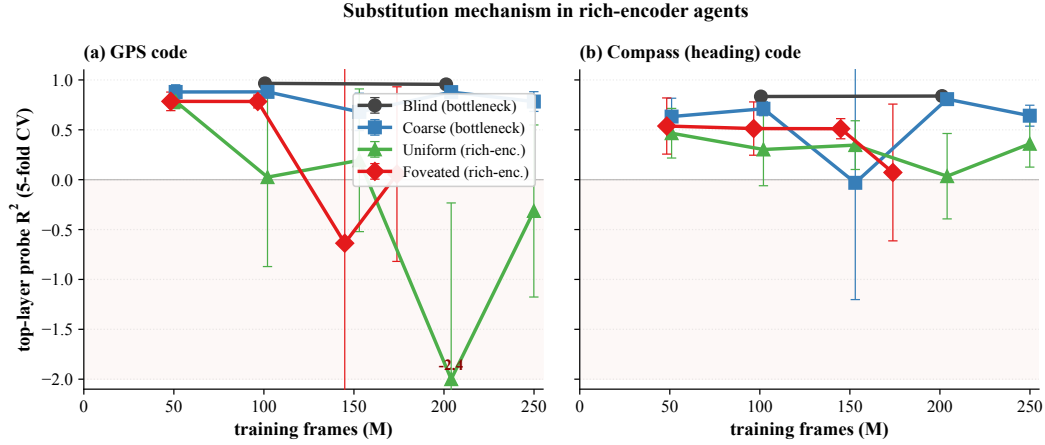


Figure 26: Cross-training probe profile across 4–5 training checkpoints per condition, truncated at 250M frames. (a) Top-layer GPS R^2 . (b) Top-layer compass R^2 . (500 ep/probe, 5-fold CV, single seed; hollow dotted = partial-data ckpts; negative- R^2 outliers clipped at -2.0 .)

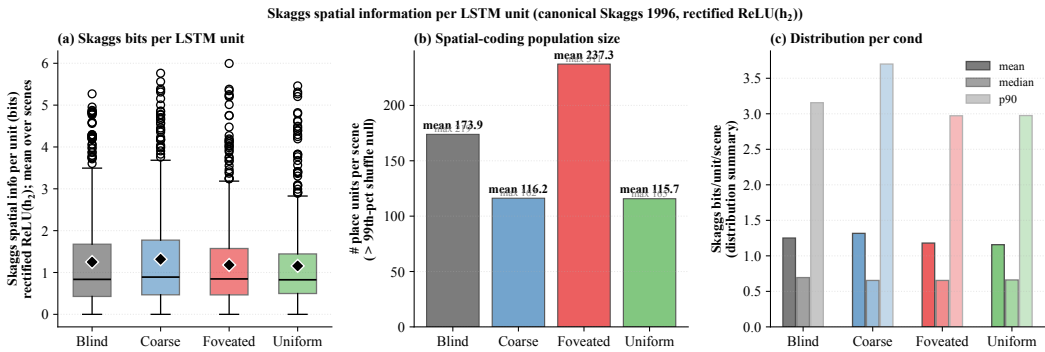


Figure 27: **Skaggs spatial information per LSTM unit.** (a) Distribution of per-unit Skaggs bits (averaged across 15 top scenes per condition); bottleneck and rich-encoder conditions overlap substantially (means ≈ 1.16 – 1.32 bits). (b) Number of place units per scene exceeding the 99th percentile of a 100-shuffle null; foveated has the largest count (≈ 237 /scene), blind/coarse/uniform ≈ 116 – 174 . (c) Distribution summary; cross-condition differences sit within the within-condition spread. The canonical Skaggs metric on rectified activations does not by itself discriminate bottleneck from rich-encoder regimes.

Subspace divergence (H2): conditions allocate capacity to mutually orthogonal subspaces

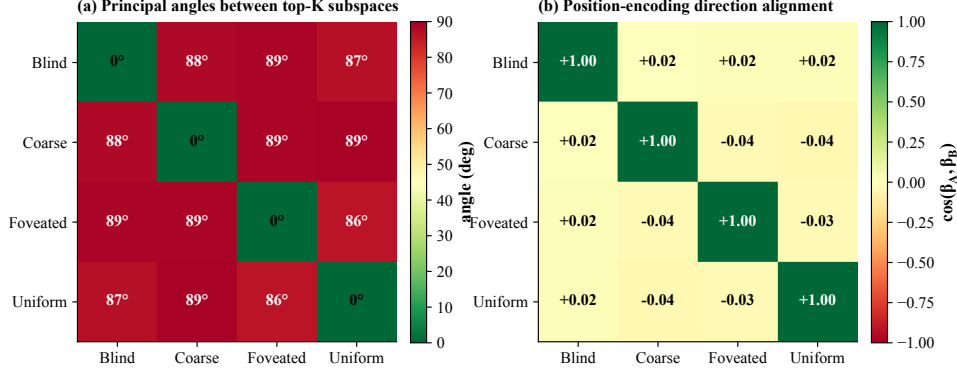


Figure 28: **Subspace divergence, quantitative form.** (a) Mean principal angles between top- K PCA subspaces of $\mathbf{h}_2$ ($K \in [4, 33]$ per condition, covering $\sim 90\%$ variance; 20 000 episode-step samples per condition). Off-diagonal angles in $[69, 87]$ (most pairs in $[77, 87]$) — conditions occupy mostly-orthogonal subspaces; the closest pair (foveated–foveated-logpolar, 69) is the same-family rich-encoder pair. (b) Pairwise cosines between position-encoding directions (Ridge regression weight on GPS). All off-diagonal cosines in $[-0.094, +0.102]$ — position is encoded along essentially uncorrelated directions across conditions. Single seed.

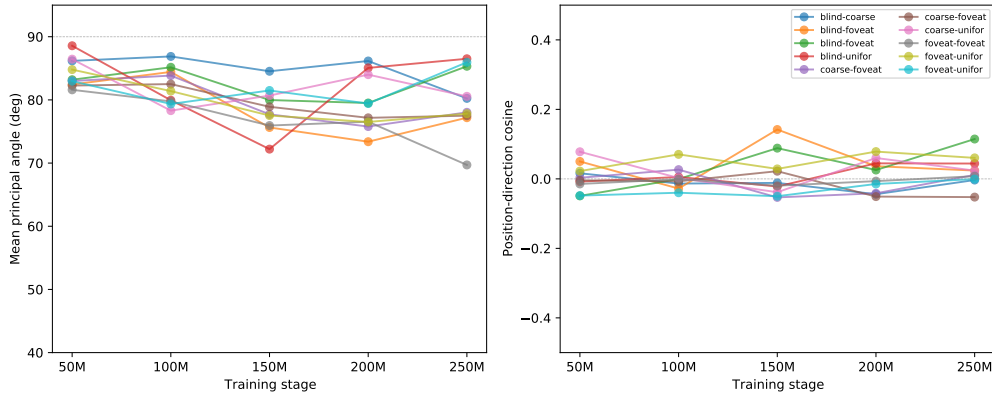


Figure 29: **Subspace evolution over training.** (a) Mean principal angle between condition-pair top- K subspaces, plotted across five training stages ($\sim 50\text{M}$ to $\sim 250\text{M}$ frames; sighted at ckpt 10/20/30/40/49, blind at 10/15/20/25). Angles are already $[82, 89]$ at the earliest probed stage and stay near-orthogonal throughout. (b) Position-encoding direction cosines are essentially zero throughout (range $[-0.05, +0.14]$), indicating direction divergence is committed before the earliest probed checkpoint. Single seed; 4–5 ckpts per condition.

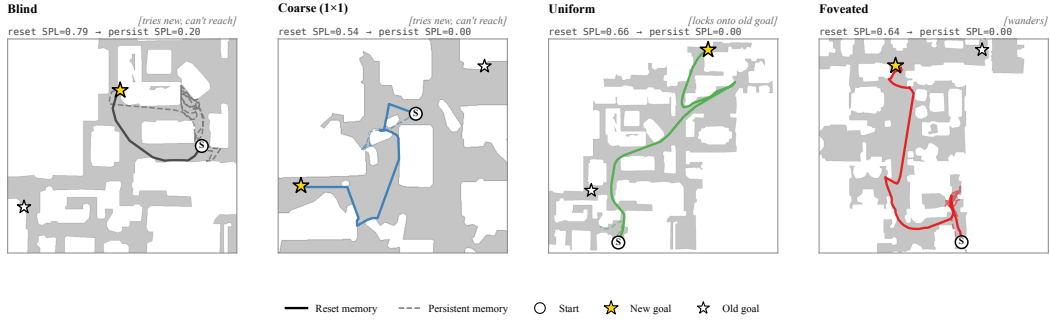


Figure 30: Persistent-memory failure modes — one canonical paired-episode case per condition (extended catalog: Appendix E.6). Aggregate per-condition margin = avg dist(persistent end → old goal) − avg dist(persistent end → new goal) over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\text{m}$): Blind −0.38m ($n=27$), Coarse −0.57m ($n=35$), Foveated −0.59m ($n=16$), **Uniform +1.83m ($n=46$, closer to OLD goal)**. Single seed.